DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

QUALITY ASSURANCE

Question Bank with Model Answers

Course Title	Quality Assurance	**Course Code**	IM353IA
Semester	V	**Units Covered**	Unit I and Unit II
Compiled From	CIE and SEE papers, 2019–2026	**Questions**	Part A: 58 Part B: 50

Notes:

1. Questions are compiled from the CIE and SEE question papers of 2019–2026. The unit and the source paper are shown below each question.
2. Part A contains fill-in-the-blank and short-answer questions. Part B contains descriptive and numerical questions (4 to 16 marks).
3. Model answers are written to the depth expected for the marks allotted. Numerical answers show the complete working, and the final result is stated after **Answer:**.
4. Where a question repeats an earlier one, the answer refers to the earlier question (the reference is a link).
5. M: Marks; BT: Bloom's Taxonomy level; CO: Course Outcome. Control chart factors (A_2, A_3, B_3, B_4, D_3, D_4, d_2, c_4) are taken from the standard table of factors for variables control charts. Answers follow the course lecture slides (Quality Improvement in the Modern Business Environment, Methods and Philosophy of Statistical Process Control, the Quality 1.0 to 4.0 guide and the Unit II presentation on control charts and process capability); topics not covered in those slides are answered from the prescribed textbook.

Sl. No.	Questions	M	BT	CO
	PART – A			
1	Write different definitions of quality. Source: Unit I | SEE Nov/Dec 2019 (16IM53), Q1.1; SEE Mar/Apr 2023 (18IM53), Q1.1 (any one definition)	02	–	–

- **Traditional definition:** quality means fitness for use. It has two aspects: quality of design (the intentional differences in grade or level of quality, for example materials, features and specifications) and quality of conformance (how well the product conforms to the specifications required by the design).
- **Modern definition:** quality is inversely proportional to variability. When the variability in the important characteristics of a product decreases, its quality increases. In the transmission example, the transmissions with less variability about the target had far lower warranty cost.
- **Quality improvement** is therefore the reduction of variability in processes and products. An equivalent definition, useful in service or transactional businesses, is the elimination of waste.
- **Dimensions:** quality is multifaceted and is described by eight dimensions: performance, reliability, durability, serviceability, aesthetics, features, perceived quality and conformance to standards.

2	List any four dimensions of quality. Source: Unit I | SEE Feb/Mar 2025 (IM353IA), Q1.5	02	1	1

Answer: Any four of Garvin's eight dimensions: performance, reliability, durability, serviceability, aesthetics, features, perceived quality and conformance to standards.

Example: for a car, performance is acceleration and mileage, reliability is how seldom it needs repair, durability is its service life and serviceability is how quickly and cheaply it can be repaired.

3	The _____ are the means of translating the term 'fitness for use' into the language of technologists. Source: Unit I | SEE Fast Track Jul 2019 (16IM53), Q1.1	01	–	–

Answer: quality characteristics

Quality characteristics (critical-to-quality or CTQ characteristics) are the measurable parameters, such as length, weight, taste or reliability, that jointly describe what the user thinks of as quality.

4	A physical or chemical property or any other requirement used to define the nature of the product is termed as _____. Source: Unit I | SEE Jan/Feb 2021 (16IM53), Q1.2	01	–	–

Answer: a quality characteristic

Quality characteristics may be physical (length, weight, voltage, viscosity), sensory (taste, appearance, colour) or time-oriented (reliability, durability, serviceability).

5	Quality cannot be inspected into a product nor can it be advertised in; it must be designed and built in. Comment on the statement. Source: Unit I | SEE Jan/Feb 2021 (16IM53), Q1.1	02	–	–

The statement is correct. Quality means reduced variability, and variability has to be removed from the design and the process; it cannot be sorted out or advertised afterwards.

- **Inspection** only separates good units from bad after they are made; the cost of making the defective units has already been incurred, and mass inspection is never completely effective. This is why Deming's third point is to cease reliance on mass inspection to improve quality.
- **Advertising** can raise perceived quality but cannot change the variability of the product itself.
- **Design and process:** quality of design (right materials, specifications and processes) and quality of conformance (capable processes kept in statistical control with control charts, and key factors found with designed experiments) build quality into the product. Deming's fifth point is to improve constantly and forever the system of production and service.

6	Differentiate between defect and defective. Source: Unit I | CIE-I, 26/11/2024; CIE-I, 07/11/2025; SEE Jan/Feb 2021 (16IM53), Q1.7	02	2	1

Basis	Defect (nonconformity)	Defective (nonconforming product)
Meaning	A particular failure to meet a specification or requirement. One unit may contain several defects.	A unit of product that fails to meet one or more of its specifications.
Relation	Not all products containing a defect are necessarily defective: a minor nonconformity may not affect the safe or effective use of the product.	A unit is defective when its nonconformities are serious enough to affect use.
Counted as	Number of defects per unit.	Number or fraction of defective units in a sample.

Chart used	c chart and u chart	p chart and np chart
Example	A scratch or a missing rivet on a car door	The car door that has the scratch

7	Consider the process of costing (casting). Give an example of a defect. Source: Unit I | SEE Mar/Apr 2023 (18IM53), Q1.7	01	–	–

For a casting process, typical defects are blow holes, porosity, shrinkage cavities, cold shuts, misruns, hot tears and dimensional mismatch. **Example:** a blow hole on the machined face of a cast pump housing.

8	List the statistical methods for quality control and improvement. Source: Unit I | CIE-I, 07/11/2025	02	1	1

The three major areas of statistical methods for quality control and improvement are:

(1) **Statistical process control (SPC):** control charts plus other problem-solving tools; used to monitor processes and to reduce variability by eliminating assignable causes. It is an on-line technique.

(2) **Designed experiments (DOX):** systematically varying the controllable factors, for example in a factorial experiment, to discover the key factors that influence process performance and to optimize the process. It is an off-line technique.

(3) **Acceptance sampling:** inspection of a sample from a lot (outgoing or incoming inspection) and a decision on the disposition of the lot: accept and ship, or reject and scrap or rework.

Historically acceptance sampling came first, then process control, and designed experiments were adopted most recently.

9	What is the primary goal of quality assurance? Source: Unit I | CIE-I, 26/11/2024; CIE-I, 07/11/2025	02	2	1

The primary goal of quality assurance is to ensure that the quality levels of products and services are properly maintained and that supplier and customer quality issues are properly resolved. It is achieved through a documented quality system with four components: policy (what is to be done and why), procedures (the methods and personnel that implement the policy), work instructions and specifications (product-, department-, tool- or machine-oriented) and records (evidence that the policies, procedures and work instructions were followed, used for traceability, customer complaints, corrective action and recalls).

10	What is the first step in Quality Assurance? Source: Unit I | SEE Jan/Feb 2021 (18IM53), Q1.1	01	–	–

The first step is **quality planning**: identifying the customers (external and internal) and their needs (listening to the voice of the customer), and then setting the quality policy and objectives that the quality assurance system will document and maintain.

11	Differentiate between Quality Control (QC) and Quality Assurance (QA). Source: Unit I | SEE Feb/Mar 2025 (IM353IA), Q1.1	02	2	1

Basis	**Quality Control and Improvement (QC)**	**Quality Assurance (QA)**
Meaning	The set of activities used to ensure that products and services meet requirements and are improved on a continuous basis.	The set of activities that ensures the quality levels of products and services are properly maintained and that supplier and customer quality issues are properly resolved.
Main tools	Statistical techniques: SPC and designed experiments, since variability is a major source of poor quality.	Documented quality system: policy, procedures, work instructions and specifications, and records.
Approach	Project-by-project improvement by teams with specialized knowledge of statistical methods; projects have significant business impact.	System oriented; document control, so that the documented system is followed and problems are prevented.

12	What are the main objectives of a quality audit? Source: Unit I | SEE Feb/Mar 2025 (IM353IA), Q1.4	02	2	1

- Verify that the documented quality system (policy, procedures, work instructions and specifications, records) is being followed.
- Check the present quality management system against the requirements of the standard, as in the initial (phase I) audit before certification.
- Identify nonconformities and opportunities for improvement and start corrective action.
- Support the certification audit by the registrar and the periodic (usually annual) surveillance audits that confirm the system continues to meet the standard.

13	How does ISO certification benefit organizations? Source: Unit I | CIE-I, 26/11/2024	01	1	2

ISO certification demonstrates a supplier's ability to control its processes. It gives a documented, consistent quality management system; satisfies customers who require certified suppliers (often a condition for doing business, nationally and internationally); clarifies responsibilities; and builds continual improvement into the organization through internal audits and management review.

14	Write any four benefits of ISO 9000. Source: Unit I | SEE Nov/Dec 2019 (16IM53), Q1.3	02	–	–

- Documented and consistent processes, so products are made the same way every time.
- Greater customer confidence and access to customers and markets that require ISO 9000 certification.
- Clear responsibilities and authority, and better training of personnel.
- Continual improvement through internal audits, corrective action and management review, which reduces waste and cost.

15	What is the broad purview of ISO 9000 standards? Source: Unit I | SEE Mar/Apr 2023 (18IM53), Q1.2	01	–	–

The International Organization for Standardization (ISO, founded in 1946 in Geneva) issued its first quality system standards in 1987. ISO 9000 is a generic standard, broadly applicable to any type of organization, and is often used to demonstrate a supplier's ability to control its processes. The ISO 9000 series has three standards: ISO 9000:2000 Quality Management Systems, fundamentals and vocabulary; ISO 9001:2000 Quality Management Systems, requirements; and ISO 9004:2000 Quality Management Systems, guidelines for performance improvement.

16	What is the primary objective of ISO 14000 standards? Source: Unit I | CIE-I, 07/11/2025; SEE Feb/Mar 2025 (IM353IA), Q1.3	02	2	1

ISO 14000 (ISO 14001) is a family of environmental management standards. Its primary objective is to help an organization establish an effective environmental management system (EMS) that identifies and controls the environmental impact of its activities, products and services, complies with environmental laws and regulations, prevents pollution and continually improves its environmental performance.

17	The concept of integrating digital technologies with QA practices is called _____. Source: Unit I | SEE Feb/Mar 2026 (IM353IA), Q1.1	01	2	2

Answer: Quality 4.0

Quality 4.0 (2010s to today, aligned with Industry 4.0) is the connected, intelligent approach to quality made possible by the technologies of the Fourth Industrial Revolution: digital technologies are integrated with quality practices so that quality is predicted and optimized in real time.

18	What is Quality 4.0? Mention two technologies used. Source: Unit I | SEE Feb/Mar 2026 (IM353IA), Q1.2	02	2	1

Quality 4.0 is the current stage of quality management (after Quality 1.0 inspection, 2.0 quality control and 3.0 quality assurance). It uses continuous streams of data, artificial intelligence and connected systems to predict problems before they happen and to optimize quality in real time, instead of relying on periodic checks or after-the-fact statistics.

Two technologies: (i) IoT-connected sensors, which stream quality-relevant data from machines, products and processes continuously; (ii) artificial intelligence and machine learning, whose predictive models flag the conditions that lead to defects before they occur. Others are big data analytics, cloud-based quality management systems, blockchain traceability, robotics and automation, and digital twins.

19	Mention four key core technologies that drive Quality 4.0. Source: Unit I | CIE-I, 26/11/2024	02	2	3

Quality 4.0 rests on six technology pillars (any four):

(1) **IoT sensors:** continuous machine and product data collection on the shop floor, replacing periodic manual measurement.
(2) **AI and machine learning:** predictive defect detection and automated root-cause analysis from historical and real-time data.
(3) **Big data analytics:** finding patterns across massive, multi-source datasets that are too complex for manual review.
(4) **Cloud-based quality management systems (QMS):** centralized quality data that is accessible and consistent across every site, team and supplier.
(5) **Blockchain traceability:** tamper-proof records of materials, processes and inspections across the supply chain.
(6) **Robotics and automation:** consistent, tireless inspection and process execution, reducing the variability introduced by manual labour.

Digital twins, virtual models of products and processes used to simulate, test and optimize quality decisions before they are implemented physically, are also counted among the core technologies.

20	The cost associated with measuring, evaluating or auditing the product comes under the costs of _____. Source: Unit I | SEE Fast Track Jul 2019 (16IM53), Q1.2	01	–	–

Answer: appraisal

Appraisal costs are the costs of measuring, evaluating or auditing products, components and purchased materials to ensure conformance to standards.

21	The costs related to formal technical reviews and test equipment reviews are related to _____ type of quality cost. Source: Unit I | SEE Jan/Feb 2021 (18IM53), Q1.11	01	–	–

Answer: prevention

Formal technical reviews (new product and design reviews), quality planning, test equipment and training are incurred to stop defects from occurring, so they are prevention costs. In the quality cost table, new products review is listed under prevention; inspection and test of product and the maintenance of the accuracy of test equipment in routine inspection are appraisal costs.

22	List the 7 QC tools. Source: Unit I | CIE-I, 26/11/2024; SEE Nov/Dec 2019 (16IM53), Q1.4	02	1	3

The seven major SPC problem-solving tools (the magnificent seven) are:

(1) Histogram or stem-and-leaf plot
(2) Check sheet
(3) Pareto chart
(4) Cause-and-effect diagram
(5) Defect concentration diagram
(6) Scatter diagram
(7) Control chart

23	State the central limit theorem. Source: Unit I | SEE Jan/Feb 2021 (16IM53), Q1.3; SEE Fast Track Jul 2019 (16IM53), Q1.3	01	–	–

If x1, x2, ..., xn are independent random variables with means μ and variances σ^2, then the distribution of their sum, and hence of the sample mean, approaches the normal distribution as n becomes large, regardless of the distributions of the individual variables. The sample mean is then approximately normal with mean μ and standard deviation $\sigma/\sqrt{n}$; this is the basis of the control limits on the $\bar{x}$ chart.

24	Differentiate between chance and assignable causes of variation. Source: Unit I | SEE Feb/Mar 2025 (IM353IA), Q1.2; SEE Jan/Feb 2021 (16IM53), Q1.4; SEE Fast Track Oct 2020 (16IM53), Q1.1	02	2	1

Basis	**Chance (common) causes**	**Assignable (special) causes**
Nature	Natural, inherent variability: the cumulative effect of many small, essentially unavoidable causes (“background noise”).	Large variability that is not part of the chance-cause pattern.
Sources	Part of the process itself; a stable system of chance causes.	Improperly adjusted machines, operator errors, defective raw material.
Process state	Process is in statistical control.	Process is out of control.
Action	Reduced only by changing the process (management action).	Detected by control charts and removed by corrective action.

25	Differences among materials, workers and machines is _____ type of cause of variation. Source: Unit I | SEE Fast Track Jul 2019 (16IM53), Q1.4	01	–	–

Answer: assignable

Differences among materials, workers and machines (defective raw material, operator errors, improperly adjusted machines) are the three typical sources of assignable causes.

26	Consider an operation on a CNC machine. Give an example of a possible assignable cause. Source: Unit I | SEE Mar/Apr 2023 (18IM53), Q1.3	01	–	–

Tool wear or a broken cutting tool, a wrong tool offset entered by the operator, loose work holding, or a batch of material with different hardness. Each produces a shift or trend in the machined dimension.

27	Statistical process control helps identify _____ type of causes that affect the process. Source: Unit I | SEE Jan/Feb 2021 (18IM53), Q1.2	01	–	–

Answer: assignable (special)

28	A control chart contains a centre line, _____ and _____. Source: Unit I | SEE Feb/Mar 2026 (IM353IA), Q1.3	01	1	1

Answer: upper control limit (UCL) and lower control limit (LCL)

A point that plots within the control limits indicates that the process is in control; a point outside the limits is evidence that the process is out of control. There is a close connection between control charts and hypothesis testing.

29	In control charts, the dividing lines between random and nonrandom deviations from the mean of the distribution are known as _____. Source: Unit I | SEE Jan/Feb 2021 (18IM53), Q1.8	01	–	–

Answer: control limits

30	The mean of the sampling distribution is equal to _____ of the process distribution. Source: Unit I | SEE Jan/Feb 2021 (18IM53), Q1.6	01	–	–

Answer: the mean

$E(\bar{x}) = \mu$; the standard deviation of the sampling distribution is $\sigma/\sqrt{n}$.

31	The percent of the sample means that will have values within ± 3 standard deviations of the distribution mean is _____. (What proportion of the normal distribution is covered under the 3σ control limits?) Source: Unit I | SEE Jan/Feb 2021 (18IM53), Q1.7; SEE Mar/Apr 2023 (18IM53), Q1.4	01	–	–

Answer: 99.73%

With three-sigma limits the probability of a point falling outside the limits when the process is in control (type I error) is 0.0027.

32	Define the term warning limit. Source: Unit I | SEE Mar/Apr 2023 (18IM53), Q1.5	01	–	–

Warning limits are inner limits, typically placed at two-sigma from the center line, inside the three-sigma control limits. In the flow-width example (center line 1.5, $\sigma_{\bar{x}} = 0.0671$) the control limits are 1.2987 and 1.7013, and the warning limits are 1.3658 and 1.6342. A point between a warning limit and a control limit is a warning that the process may be changing, so sampling is made more frequent or the sample size is increased.

33	What is ATS and ARL? Source: Unit I | CIE-I, 26/11/2024	02	1	2

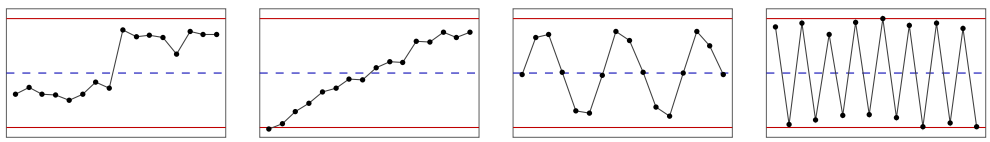
ARL (average run length): the average number of points that must be plotted before a point indicates an out-of-control condition; ARL = 1/p, where p is the probability that a point exceeds the control limits. For three-sigma limits, in-control ARL_0 = 1/0.0027 = 370.

ATS (average time to signal): the average time before a signal occurs when samples are taken h hours apart; ATS = ARL × h.

34	State any two principles of rational subgrouping. Source: Unit I | SEE Fast Track Oct 2020 (16IM53), Q1.2	02	–	–

- Subgroups should be selected so that if assignable causes are present, the chance for differences **between** subgroups is maximized, while the chance for differences due to assignable causes **within** a subgroup is minimized.
- There are two general approaches: (i) each sample consists of units produced at the same time (consecutive units), and (ii) each sample represents all units produced since the last sample, often used for decisions on the acceptance of product. Care must be taken because any process can be made to appear in statistical control just by stretching the interval between observations.

35	Sketch the extreme variation patterns of assignable cause. Source: Unit I | SEE Fast Track Oct 2020 (16IM53), Q1.3	02	–	–



(a) Sudden shift in level (b) Trend (gradual drift) (c) Cycles (d) Mixture / freaks

Figure 1: Typical nonrandom patterns caused by assignable causes

An assignable cause pushes the process out of control in one of three extreme ways: the process mean moves up, the process mean moves down, or the spread increases ($\sigma_1 > \sigma_0$), singly or together. On the chart these appear as sudden shifts in level, trends, cycles, mixtures and long runs on one side of the center line, even when all the points are inside the control limits.

36	_____ plans detail the action to be taken once an out-of-control situation is detected. Source: Unit I | SEE Jan/Feb 2021 (16IM53), Q1.6	01	–	–

Answer: Out-of-control action plans (OCAP)

An OCAP is a flow chart that gives the checkpoints (the potential assignable causes to examine, for example whether the data were entered correctly or the settings were correct) and the terminators (the action to take, such as adjusting the process or contacting process engineering) that follow an out-of-control signal. It supports the cycle of detecting the assignable cause, identifying the root cause, implementing corrective action, and verifying and following up.

37	Differentiate between attribute control chart and variable control chart. Source: Unit II | CIE-II, 07/01/2025	02	L2	CO1

Basis	**Variable control chart**	**Attribute control chart**
Data	A quality characteristic measured on a continuous numerical scale (continuous data on the degree of conformance to specifications).	A characteristic classified simply as conforming or nonconforming (present or absent), or counted as a number of defects; discrete data.
Examples	Diameter (mm), weight (g), tensile strength (MPa), time (s).	Pass/fail, go/no-go gauge result, number of scratches on a panel, an order complete or incomplete.
Charts	$\bar{x}$–R, $\bar{x}$–s, individuals and moving range (X–MR).	p, np, 100p, c and u charts.
Information	More information per unit measured, so smaller samples are enough.	Often cheaper and faster to collect, but less information-rich: a larger sample is needed for the same information.

38	When do you use attribute control charts? Source: Unit II | SEE Nov/Dec 2019 (16IM53), Q1.7	02	–	–

Attribute control charts are used when the quality characteristic is either present or absent in the product or service, so it cannot be measured on a continuous scale: a unit is classified as conforming or nonconforming (p and np charts), or the number of nonconformities is counted (c and u charts), for example an order that is complete or incomplete, or an invoice with one, two or more errors. They are the usual choice in service industries, where most quality characteristics are attributes, and wherever go/no-go gauging is used.

39	The number of invoices that contain errors is an example of _____ data. Source: Unit II | SEE Jan/Feb 2021 (16IM53), Q1.9	01	–	–

Answer: attribute

An invoice can have one, two or more errors; such counts are discrete attribute data.

40	What does an R-chart measure, and why is it important in statistical process control? Source: Unit II | CIE-II, 07/01/2025	02	L1	CO2

The R chart plots the range of each subgroup and measures the **variability (dispersion)** of the process. It is important because the $\bar{x}$ chart limits are calculated from $\bar{R}$; if the variability is not in control the $\bar{x}$ limits are meaningless. The R chart signals any gain or loss in process uniformity.

41	The usual purpose of an R-chart is to signal whether there has been a gain or loss in _____. Source: Unit II | SEE Jan/Feb 2021 (16IM53), Q1.5; SEE Fast Track Jul 2019 (16IM53), Q1.5	01	–	–

Answer: uniformity (process dispersion or variability)

42	Control limits for the R chart are _____ and _____. Source: Unit II | SEE Fast Track Oct 2020 (16IM53), Q1.4	02	–	–

Answer: UCL = $D_4\bar{R}$ and LCL = $D_3\bar{R}$ (center line $\bar{R}$)

43	The factors A_1, A_2, D_1, D_2, D_3 and D_4 used to calculate control limits depend upon _____. Source: Unit II | SEE Fast Track Jul 2019 (16IM53), Q1.7	01	–	–

Answer: the sample (subgroup) size n

44	Central tendency of a process is monitored in _____ control chart. Source: Unit II | SEE Jan/Feb 2021 (18IM53), Q1.9	01	–	–

Answer: $\bar{x}$ (X-bar)

45	Jars of pickles are sampled and weighed. Sample measures are plotted on control charts. The ideal weight should be precisely 11 oz. Which type of chart(s) would you recommend? Source: Unit II | SEE Fast Track Jul 2019 (16IM53), Q1.6	01	–	–

Weight is a variable measured on a continuous scale with a target of 11 oz, so $\bar{x}$ **and R charts** are recommended: the $\bar{x}$ chart monitors the mean fill weight and the R chart monitors the variability.

46	Write the steps followed in constructing the individual chart. Source: Unit II | CIE-I, 07/11/2025	02	2	2

(1) Collect individual measurements x_1, x_2, … (n = 1) in time order.
(2) Compute the moving ranges $MR_i = |x_i - x_{i-1}|$ and their average $\overline{MR}$; compute $\bar{x}$.
(3) Control limits: $\bar{x} \pm 3\,\overline{MR}/d_2 = \bar{x} \pm 2.66\,\overline{MR}$ (d_2 = 1.128); MR chart: UCL = 3.267 $\overline{MR}$, LCL = 0.
(4) Plot the individual values and moving ranges and interpret for points outside limits and patterns.

47	Mention any one of the uses of the chart for individual measurements. Source: Unit II | SEE Mar/Apr 2023 (18IM53), Q1.9	01	–	–

The chart for individual measurements (X–MR chart) is used when only one observation per sampling period is available or practical, so subgrouping is not possible: destructive testing (only one unit can be tested per batch), a very low production rate (for example one batch per day), homogeneous or bulk processes such as chemical batches and tank fills, and automated 100% inspection where every unit is measured individually.

48	When do you prefer to use a group control chart for monitoring the process quality? Source: Unit II | SEE Mar/Apr 2023 (18IM53), Q1.10	01	–	–

A group control chart is preferred when a process has several similar, parallel streams (multiple spindles, cavities or heads) and a separate chart for each would be unwieldy. At each sampling time one value (for example the mean) is obtained from each of the k streams, and only the highest and lowest values are plotted, each labelled with its stream number, against common control limits. It greatly reduces the number of charts, and an out-of-control signal identifies the stream to investigate; its caution is that only the extreme values are seen, so patterns in the other streams can be missed.

49	What is the key difference between a p-chart and an np-chart? Source: Unit II | CIE-II, 07/01/2025	02	L2	CO3

Basis	**p chart**	**np chart**
Plots	Fraction nonconforming $\hat{p} = D/n$ in each sample.	Actual number (count) of nonconforming units D = n$\hat{p}$.
Sample size	May vary from sample to sample (limits form a staircase).	Must be constant.
Preferred when	Samples differ in size; management reporting (the 100p chart plots the same fraction as a percentage).	Shop-floor operators read the chart directly as a headcount of bad parts; no fraction calculation needed.

50	Explain the purpose of a p-chart. Source: Unit II | CIE-II, 16/12/2025	02	3	1

The p chart tracks the proportion (fraction) of nonconforming units in each sample, on the basis of the binomial distribution. Each unit is classified simply as conforming or nonconforming; the number or type of defects on a nonconforming unit is not counted. Points are plotted against $\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n}$ (a negative LCL is set to 0), and a point outside the limits signals an assignable cause. It is widely used for final inspection results, audit results and service-quality pass/fail metrics.

51	What can you conclude when a point falls below the lower control limit on a control chart for fraction nonconforming (p-chart)? Source: Unit II | SEE Mar/Apr 2023 (18IM53), Q1.6	01	–	–

A point below the LCL does not necessarily mean improvement. It may indicate a real improvement in quality, but often it is due to an error in inspection (poorly trained inspector, wrong gauge) or in recording the data. The cause should be investigated; if it is a genuine improvement it should be made permanent.

52	The local newspaper receives several complaints per day about typographic errors. Over a seven-day period, the publisher has received calls from readers reporting the following number of errors: 4, 3, 2, 6, 7, 3 and 9. Based on these data alone, what type of control chart(s) should the publisher use? Source: Unit II | SEE Jan/Feb 2021 (16IM53), Q1.8; SEE Fast Track Jul 2019 (16IM53), Q1.8	02	–	–

Answer: a c chart (number of errors, constant inspection unit: one day's paper)

$\bar{c}$ = 34/7 = 4.857; UCL = 4.857 + 3$\sqrt{4.857}$ = 11.47, LCL = 0.

53	The control chart used for the number of defects per unit is _____. Source: Unit II | SEE Jan/Feb 2021 (18IM53), Q1.10	01	–	–

Answer: u chart

54	In a lot of 50 parts, 4 parts were rejected because of 12 non-conformities found in them. The proportion defective is _____ and the average number of non-conformities per part in the lot is _____. Source: Unit II | SEE Fast Track Jul 2019 (16IM53), Q1.9	02	–	–

Answer: proportion defective = 4/50 = 0.08; nonconformities per part = 12/50 = 0.24

55	Define C_p. How is it interpreted? Source: Unit II | SEE Feb/Mar 2026 (IM353IA), Q1.5	02	2	2

The process capability index C_p = (USL − LSL)/6σ compares the width of the specification tolerance with the natural spread (6σ) of the process, where σ is estimated from $\bar{R}/d_2$ or $\bar{s}/c_4$. It assumes the process is centred on the target, so it measures **potential** capability only.

Value	Interpretation
$C_p < 1.00$	Process spread is wider than the tolerance: not capable
$C_p = 1.00$	Process spread exactly equals the tolerance: marginal
$1.00 \leq C_p < 1.33$	Capable, but with limited margin for drift
$C_p \geq 1.33$	Generally regarded as capable (common industry minimum)
$C_p \geq 2.00$	Six Sigma level capability

56	Define C_{pk}. How is it different from C_p? Source: Unit II | SEE Feb/Mar 2026 (IM353IA), Q1.4	02	2	2

C_{pk} = min(CPU, CPL), where CPU = (USL − μ)/3σ and CPL = (μ − LSL)/3σ, with μ estimated as $\bar{\bar{x}}$. It uses whichever specification limit is closer to the process mean, so it accounts for process centring and gives the **actual**, realised capability.

- C_p ignores where the mean sits and shows potential capability if the process were perfectly centred.
- $C_{pk} \leq C_p$ always; equality holds only when the mean is exactly centred between the specification limits.
- A large gap between C_p and C_{pk} signals a centring problem: the process is capable in spread but running off-target. $C_{pk} < 0$ means the mean has moved outside the specification limits.

57	Write down the formula for the process performance index C_{pk} taking into account process centering. Source: Unit II | SEE Mar/Apr 2023 (18IM53), Q1.8; SEE Fast Track Oct 2020 (16IM53), Q1.8	02	–	–

$$C_{pk} = \min\left(\frac{USL - \mu}{3\sigma},\ \frac{\mu - LSL}{3\sigma}\right)$$

58	In Six Sigma, the value of C_p = _____. Source: Unit II | SEE Jan/Feb 2021 (18IM53), Q1.3	01	–	–

Answer: 2.0

A six-sigma process has its specification limits at $\pm 6\sigma$, so $C_p = 12\sigma/6\sigma = 2$; a C_p of 2.00 or more is described as Six Sigma level capability.

Sl. No.	Questions	M	BT	CO
	PART – B			
1	Identify the various dimensions of quality as proposed by Garvin. Explain their significance with respect to product quality, taking suitable examples. (List and explain the eight dimensions of quality.) Source: Unit I | SEE Jan/Feb 2021 (16IM53), Q2a; SEE Nov/Dec 2019 (16IM53), Q2a; SEE Fast Track Oct 2020 (16IM53), Q2a (discuss any two); SEE Feb/Mar 2026 (IM353IA), Q2a (any four)	08	2	3

Garvin (1987) described eight components or dimensions of quality. Quality is a multifaceted entity, and it is important to differentiate these dimensions when a product or service is described and evaluated.

Dimension	Question it answers	Meaning and example
1. Performance	Will the product do the intended job?	How well the product performs its specific functions; e.g. execution speed of spreadsheet software, mileage and acceleration of a car.
2. Reliability	How often does the product fail?	Complex products (appliances, automobiles, airplanes) need some repair; a car that needs frequent repair is unreliable.
3. Durability	How long does the product last?	Effective service life; very important for automobiles and major appliances.
4. Serviceability	How easy is it to repair the product?	How quickly and economically repair or maintenance is done; e.g. how long a credit card company takes to correct a billing error.
5. Aesthetics	What does the product look like?	Visual and sensory appeal: style, colour, shape, packaging; soft-drink makers use packaging to differentiate their product.
6. Features	What does the product do?	Added features beyond basic performance; e.g. spreadsheet software with built-in statistical analysis.
7. Perceived quality	What is the reputation of the company or its product?	Customers rely on past reputation; e.g. preferring an airline that is always on time and never loses luggage.
8. Conformance to standards	Is the product made exactly as the designer intended?	Meeting the requirements placed on it; e.g. a car hood fitting flush with the fender with equal gaps on all sides.

Significance

- The dimensions translate the customer's idea of quality into specific characteristics that can be designed, measured and improved.
- Quality planning uses them to develop products and services that meet or exceed customer expectations (voice of the customer).
- A firm can compete on a few chosen dimensions (for example, reliability for a car maker, aesthetics for a beverage maker) instead of all of them.
- Failures that are visible to the public lower perceived quality, so conformance and reliability directly affect reputation and repeat business.
- In implementing quality improvement the eight dimensions form a strategic management process; suppliers and the supply chain must be involved, and all three components (quality planning, quality assurance, and quality control and improvement) must be addressed.

2	Choose a specific product (e.g., a smartphone, automobile, or any consumer electronics) and analyze how the Eight Dimensions of Quality influence its acceptance by customers. Break down each dimension—such as performance, features, reliability, and serviceability—and evaluate their relative importance in shaping customer perceptions. How do these dimensions interact with each other to create a competitive advantage for the product in the market? Use relevant examples to support your analysis. Source: Unit I | CIE-I, 26/11/2024; SEE Mar/Apr 2023 (18IM53), Q2a (take a simple product and identify the 8 dimensions)	10	4	4

Product chosen: a smartphone. Each of Garvin's eight dimensions influences whether customers accept the product.

Dimension	How it shapes customer acceptance (smartphone)	Relative importance
Performance	Processor speed, camera quality, battery life, network reception and display quality decide whether the phone does its job well.	Very high: main reason for purchase
Features	Fast charging, multiple cameras, face unlock, 5G, water resistance, AI assistants differentiate it from competitors.	High: drives upgrades
Reliability	Freedom from hangs, battery failures, overheating and app crashes during use.	High: failures cause returns
Durability	Number of years the phone remains usable: glass strength, battery ageing, years of software updates.	Medium to high
Serviceability	Ease, speed and cost of repair: service-centre network, spare-part availability, warranty handling.	Medium
Aesthetics	Design, colour, thinness, materials (glass, metal) and packaging; a strong reason for premium buyers.	Medium to high
Perceived quality	Brand reputation, reviews and past recalls (for example, battery-fire recalls damage the brand).	High: affects loyalty
Conformance	Every unit built exactly to design: no gaps in the body, display without dead pixels, specified battery capacity.	Basic requirement

Interaction of the dimensions and competitive advantage

- Performance and features attract the first purchase, but reliability, durability and serviceability decide satisfaction over the phone's life and hence repeat purchase.
- Conformance supports reliability: tighter manufacturing variability gives fewer early failures, which builds perceived quality.
- Aesthetics and perceived quality allow a price premium; a strong brand can win even with similar performance.
- Trade-offs exist: more features and a thinner body (aesthetics) can reduce battery life (performance) and durability, so a firm chooses the combination its target segment values most.
- A product that is strong on the dimensions its customers rank highest, and adequate on the rest, gains a competitive advantage.

3	Analyze how the Eight Dimensions of Quality can be used to evaluate the performance of a consumer product. Examine each dimension's effect on customer satisfaction and illustrate your analysis with a detailed case study from the consumer goods or electronics industry. Source: Unit I | CIE-I, 07/11/2025	10	4	3

The eight dimensions provide a checklist for evaluating a consumer product, because each dimension corresponds to a question the customer asks about the product.

Case study: a front-loading washing machine

Dimension	Evaluation of the washing machine	Effect on customer satisfaction
Performance	Cleaning effectiveness, spin speed, water and energy consumption, noise.	Core satisfaction; poor cleaning means rejection.
Features	Inverter motor, steam wash, Wi-Fi app control, quick-wash programmes.	Delight factor; justifies a higher price.
Reliability	Frequency of breakdowns of the drum bearing, pump and control board.	Each failure causes strong dissatisfaction.
Durability	Expected service life (10 years or more), motor warranty.	Long life reinforces value for money.
Serviceability	Availability of service engineers, time to repair, cost of parts.	Fast service can recover a dissatisfied customer.
Aesthetics	Colour, door design, display panel, fit in the home.	Influences choice among similar models.
Perceived quality	Brand reputation, star energy rating, reviews.	Builds trust and loyalty.
Conformance	Each unit meets design specifications: no leaks, correct drum balance.	Expected by all customers; any lapse is noticed.

Analysis: customers rank performance, reliability and serviceability highest for appliances, so a manufacturer that reduces variability in critical components (conformance), offers long warranties (durability) and has a strong service network earns high perceived quality and repeat business. Features and aesthetics differentiate models within the same price segment.

4	Define ‘quality'. Giving two examples each, explain the five quality characteristics. Source: Unit I | SEE Fast Track Jul 2019 (16IM53), Q2a	06	–	–

Quality means fitness for use (the traditional definition, with quality of design and quality of conformance); in the modern definition, quality is inversely proportional to variability. Every product possesses a number of elements that jointly describe what the user or consumer thinks of as quality. These parameters are called quality characteristics (also critical-to-quality or CTQ characteristics). Juran classifies them into five types:

Characteristic	Meaning	Two examples
1. Technological (physical)	Measurable physical or chemical properties.	Length or diameter of a shaft; hardness of steel
2. Psychological (sensory)	Characteristics judged by the senses and taste.	Taste of food; appearance and colour of a car
3. Time-oriented	Behaviour of the product over time.	Reliability of a television; durability of a tyre
4. Contractual	Provisions guaranteed to the customer.	Warranty period of a refrigerator; free-service clauses
5. Ethical	Conduct of the people who deliver the product or service.	Courtesy of sales staff; honesty of a repair shop

In the lecture classification the same characteristics fall into three groups: physical (length, weight, voltage, viscosity), sensory (taste, appearance, colour) and time orientation (reliability, durability, serviceability). These correspond to the technological, psychological and time-oriented types above; contractual and ethical characteristics complete the five.

5	What do you mean by quality cost? Why is it so important? Differentiate between failure cost and appraisal cost. Source: Unit I | SEE Jan/Feb 2021 (16IM53), Q2b	08	–	–

Quality costs are the categories of costs associated with producing, identifying, avoiding or repairing products that do not meet requirements. They are classified as prevention, appraisal, internal failure and external failure costs.

Importance of quality costs

- The cost of quality has increased because of the increasing complexity of products associated with advances in technology.
- There is increasing awareness of life-cycle costs, including maintenance, spare parts and the cost of field failures.
- Quality engineers can communicate quality issues in the language management understands, money; quality costs become a financial control tool.
- Quality-cost analysis (often a Pareto analysis) identifies opportunities for cost reduction. Because of the leverage effect, a rupee invested in prevention and appraisal can save many rupees of failure costs.

Failure cost versus appraisal cost

Basis	Appraisal cost	Failure cost
Meaning	Costs of measuring, evaluating or auditing products, components and purchased materials to ensure conformance to standards.	Costs incurred because products, components or services fail to meet requirements.
When incurred	Before and during production, to determine the condition of the product.	Internal failure: found before delivery. External failure: found after delivery to the customer.
Examples	Incoming material inspection, product inspection and test, materials consumed in testing, calibration of test equipment.	Internal: scrap, rework, retest, downtime, yield losses, downgrading. External: complaint adjustment, returned product, warranty charges, liability costs.
Behaviour	A cost of conformance; planned.	A cost of nonconformance; would disappear if there were no defects.

6	List and explain the four (major) categories of quality cost. Source: Unit I | SEE Nov/Dec 2019 (16IM53), Q2b; SEE Fast Track Oct 2020 (16IM53), Q2b	08	–	–

Quality costs are those categories of costs associated with producing, identifying, avoiding or repairing products that do not meet requirements. Many organizations use four categories:

Category	Definition	Subcategories
Prevention costs	Costs of efforts in design and manufacturing directed toward preventing nonconformance; costs to "make it right the first time".	Quality planning and engineering, new products review, product/process design, process control, burn-in, training, quality data acquisition and analysis
Appraisal costs	Costs of measuring, evaluating or auditing products, components and materials to ensure conformance.	Inspection and test of incoming material, product inspection and test, materials and services consumed, maintaining accuracy of test equipment
Internal failure costs	Costs when products fail to meet requirements and the failure is discovered before delivery.	Scrap, rework, retest, failure analysis, downtime, yield losses, downgrading (off-specing)
External failure costs	Costs when the product does not perform satisfactorily after delivery to the customer.	Complaint adjustment, returned product/material, warranty charges, liability costs, indirect costs (loss of reputation and market share)

Quality costs may be 4 to 5 percent of sales in some organizations and 35 to 40 percent in others. Many organizations spend too much on appraisal and too little on prevention; shifting effort to prevention reduces total quality cost.

7	Classify the Cost of Good Quality and Cost of Poor (Bad) Quality in the automotive industry. Provide examples of each cost category and explain how they can be managed effectively to improve product performance and reduce overall expenses. Source: Unit I | CIE-I, 26/11/2024; SEE Feb/Mar 2025 (IM353IA), Q2b	10	2	1

Classification

Group	Category	Examples in the automotive industry
Cost of good quality (cost of conformance)	Prevention	Quality planning and APQP, design FMEA and design reviews, supplier development, poka-yoke on assembly lines, SPC on critical dimensions, operator training, burn-in and endurance testing
	Appraisal	Incoming inspection of castings and bought-out parts, in-process gauging, end-of-line testing (wheel alignment, emission, leak and road tests), calibration of gauges, audits
Cost of poor quality (cost of nonconformance)	Internal failure	Scrap of rejected engine blocks, rework of paint defects, retesting, line stoppage (downtime), yield losses, downgrading of parts
	External failure	Warranty claims, product recalls (for example, airbag or brake recalls), complaint handling, returned vehicles, product liability suits, loss of reputation and market share

How the costs can be managed effectively

- **Invest in prevention:** robust design, design of experiments, supplier quality programmes and mistake-proofing; a rupee spent in prevention may save ten to a hundred rupees of failure cost (leverage effect).
- **Use SPC to reduce variability:** control charts on critical processes reduce scrap and rework and allow appraisal (100% inspection) to be reduced to sampling.
- **Pareto analysis of quality costs:** attack the few defect types (for example, paint defects or weld failures) that cause most of the failure cost.
- **Track quality costs** as a percentage of sales and report them to management so that improvement projects are selected on business impact.
- **Root-cause corrective action** on field failures (warranty data) prevents recalls and protects perceived quality.

As prevention increases, failure costs fall sharply and total quality cost falls, while product performance and customer satisfaction improve.

8	"Spending on prevention of poor quality leads to decrements in appraisal costs." Do you agree with this statement? Why or why not? Discuss this with the focus on components of these quality costs. Source: Unit I | SEE Fast Track Jul 2019 (16IM53), Q2b	04	–	–

Yes, generally. Prevention costs (quality planning, product/process design, process control, training) are aimed at making the product right the first time. When processes are capable and in statistical control, fewer defects are produced, so less inspection and testing is needed: 100% inspection can be replaced by sampling or by control charts, and incoming inspection is reduced when suppliers are certified. Thus appraisal costs (inspection and test, materials consumed, test equipment) fall, and internal and external failure costs fall even more.

However, appraisal cannot be reduced to zero. Some inspection, audits and calibration remain necessary to verify conformance, and before a very high quality level is reached the incremental cost of prevention rises faster than the savings. The statement holds over the normal working range: many organizations spend too much on appraisal and too little on prevention.

9	Evaluate the organizational, operational, and strategic benefits of implementing the ISO 9000 quality standard, and explain how it contributes to continuous improvement and competitive advantage. Source: Unit I | CIE-I, 07/11/2025	10	3	3

ISO 9000 is a generic standard for quality management systems, broadly applicable to any type of organization and widely used to demonstrate a supplier's ability to control its processes. The series has three standards: ISO 9000:2000 (fundamentals and vocabulary), ISO 9001:2000 (requirements) and ISO 9004:2000 (guidelines for performance improvement). ISO 9001:2000 has eight clauses: scope, normative references, definitions, quality management systems, management responsibility, resource management, product or service realization, and measurement, analysis and improvement. Clauses 4 to 8 are the most important.

Organizational benefits

- Management commitment, a quality policy and measurable quality objectives give the organization a common direction.
- Responsibilities and authorities are defined and communicated; employees are trained and competent.
- A documented system (quality manual, procedures, records) preserves knowledge and reduces dependence on individuals.

Operational benefits

- Processes are planned and carried out under controlled conditions, giving consistent output and fewer errors.
- Purchased material is controlled, and nonconforming product is identified and prevented from use.
- Records allow traceability of batches, which helps in complaints, corrective actions and recalls.
- Reduced scrap, rework and customer complaints lower operating costs.

Strategic benefits

- Certification is often a condition for supplying large customers and for exporting; industry versions such as ISO/TS 16949 for automotive and AS 9100 for aerospace build on it.
- It builds customer confidence and improves the company's image.

Certification and continuous improvement

To be certified, a company selects a registrar and prepares for a certification audit. Preparation includes an initial (phase I) audit that checks the present quality management system against the standard, teams that develop and implement the key clauses, training of personnel, documentation and any new components of the system. After certification the registrar carries out periodic surveillance audits, usually annually. Clause 8 requires monitoring of customer perception, internal audits, analysis of data, corrective and preventive action and continual improvement, and management reviews the results, so the Plan-Do-Check-Act cycle is built into the organization.

Limitation

The certification process focuses heavily on quality assurance, without sufficient weight on quality planning and on quality control and improvement. There is no single independent authority that licenses, regulates or qualifies registrars, and certification does little by itself to prevent poor quality products from being designed, manufactured and delivered (the Bridgestone/Firestone tire recall occurred at a plant certified to QS 9000 and ISO 14001). The competitive advantage therefore comes when the documented system is combined with real reduction of variability using statistical methods.

10	What are the key benefits of implementing the ISO 14000 standard for an organization? Source: Unit I | CIE-I, 26/11/2024	10	2	2

ISO 14000 is a family of international standards on environmental management; ISO 14001 specifies the requirements of an environmental management system (EMS). Key benefits:

(1) **Systematic environmental management:** identifies the environmental aspects and impacts of activities, products and services and sets objectives and targets to control them.
(2) **Legal compliance:** ensures that environmental laws and regulations are identified and met, reducing the risk of penalties and litigation.
(3) **Pollution prevention and resource efficiency:** reduced waste, emissions, energy and water use lower operating costs.
(4) **Market access and customer confidence:** many customers and export markets prefer or require ISO 14001-certified suppliers.
(5) **Improved image and stakeholder relations:** demonstrates corporate social responsibility to the public, investors and regulators.
(6) **Risk management:** better preparedness for environmental emergencies and lower insurance and liability costs.
(7) **Continual improvement:** the Plan-Do-Check-Act structure with audits and management reviews improves environmental performance over time.
(8) **Integration:** its structure is compatible with ISO 9001, so quality and environmental systems can be managed together.

11	What is the true meaning of a six sigma process? List out the percent inside specifications and the defectives in terms of PPM level for sigma limits of specifications from 1 to 6, assuming normal distribution and a perfectly centered process. What happens to these two performance measures when there is a shift in the mean of the process to the extent of $\pm 1.5\sigma$ from the target? Discuss. Source: Unit I | SEE Mar/Apr 2023 (18IM53), Q2b	08	–	–

Meaning of a six-sigma process: the process standard deviation is so small that the specification limits are six standard deviations from the mean (target) on each side, so the specification width is 12σ and $C_p = 2$. Motorola's six-sigma concept (started in 1987) recognises that a process mean does not stay exactly on target: even if the mean shifts by 1.5σ, a six-sigma process produces only 3.4 defective parts per million (ppm). Six Sigma is both this quality level and a disciplined, project-by-project programme (DMAIC: define, measure, analyze, improve, control) for reducing variability and achieving significant business impact.

Percent inside specifications and ppm defective (normal distribution)

Spec. limits	Centered: % inside	Centered: ppm defective	Mean shifted 1.5σ: % inside	Shifted: ppm defective
$\pm 1\sigma$	68.27	317,300	30.23	697,700
$\pm 2\sigma$	95.45	45,500	69.13	308,700
$\pm 3\sigma$	99.73	2,700	93.32	66,810
$\pm 4\sigma$	99.9937	63	99.3790	6,210
$\pm 5\sigma$	99.999943	0.57	99.97670	233
$\pm 6\sigma$	99.9999998	0.002	99.999660	3.4

Note: Centered process: proportion inside $= 2\Phi(k) - 1$. *Mean shifted by* 1.5σ*: proportion inside* $= \Phi(k - 1.5) - \Phi(-k - 1.5)$. *For* $\pm 2\sigma$ *shifted, 100 - 69.13 = 30.87%, i.e. 308,700 ppm (the lecture slide prints 608,700, which is a misprint).*

Effect of a 1.5σ shift

When the mean shifts by 1.5σ, the percentage inside the specifications falls and the ppm defective rises sharply at every sigma level. For a three-sigma process the defect rate rises from 2,700 ppm to 66,810 ppm, which is unacceptable for a product with many components: with 10 components, 99% good for each is not good enough. For a six-sigma process it rises only from 0.002 ppm to 3.4 ppm, so the product remains essentially defect-free. This robustness to drift is why six-sigma capability is the goal for complex products.

12	How is the application of concepts of statistical quality control advantageous over 100% inspection? Explain. Source: Unit I | SEE Jan/Feb 2021 (18IM53), Q2a	10	–	–

Acceptance sampling and statistical process control give better protection at lower cost than 100% inspection:

(1) **Lower cost:** only a sample is inspected, so inspection cost, labour and time are much lower.
(2) **Less handling damage:** fewer units are handled, so less damage occurs during inspection.
(3) **Applicable to destructive testing:** when the test destroys the item, 100% inspection is impossible.
(4) **Higher accuracy:** 100% inspection suffers from inspector fatigue and boredom, so many defectives slip through; a smaller sample is inspected more carefully.
(5) **Stronger motivation for quality:** rejection of a whole lot puts pressure on the supplier to improve the process.
(6) **Prevention instead of detection:** SPC control charts detect assignable causes as soon as they occur, so corrective action is taken before many nonconforming units are made. 100% inspection only sorts good from bad after the cost of making defectives has been incurred; quality cannot be inspected into the product.
(7) **Information for improvement:** control charts estimate process parameters and capability and give diagnostic information; 100% inspection does not reduce variability.

Sampling does carry the risk of accepting bad lots or rejecting good lots (consumer's and producer's risks), but these risks are known and controlled by the design of the plan.

13	A certain type of storage battery lasts on an average 3 years with a standard deviation of 0.5 year. Assuming normal distribution, find the percentage of batteries that last (a) less than 2 years, (b) less than 4 years, (c) between 2.5 and 4.5 years, (d) less than 4 years. Source: Unit I | CIE-I, 26/11/2024	10	3	3

Battery life $X \sim N(\mu = 3,\ \sigma = 0.5)$. Standardize with $z = (x - \mu)/\sigma$ and use the standard normal table $\Phi(z)$.

$$\text{(a) } P(X < 2) = P\left(z < \frac{2-3}{0.5}\right) = \Phi(-2.00) = 0.0228$$

$$\text{(b) } P(X < 4) = P\left(z < \frac{4-3}{0.5}\right) = \Phi(2.00) = 0.9772$$

$$\text{(c) } P(2.5 < X < 4.5) = \Phi\left(\frac{4.5-3}{0.5}\right) - \Phi\left(\frac{2.5-3}{0.5}\right)$$
$$= \Phi(3.00) - \Phi(-1.00) = 0.99865 - 0.15866 = 0.8400$$

$$\text{(d) } P(X < 4) = \Phi(2.00) = 0.9772$$

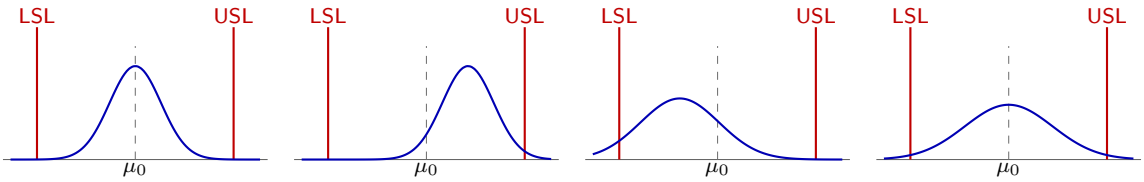
Answer: (a) 2.28% of the batteries last less than 2 years; (b) 97.72% last less than 4 years; (c) 84.00% last between 2.5 and 4.5 years; (d) 97.72% last less than 4 years (if part (d) is read as "more than 4 years", the answer is 2.28%).

14	What is variability? Distinguish clearly between chance causes and assignable causes of variation. Source: Unit I | SEE Nov/Dec 2019 (16IM53), Q3a	08	–	–

Variability: no two units of a product are ever identical. There is a certain amount of variability in every product, arising from differences in materials, in the performance and operation of the manufacturing equipment, and in the way operators perform their tasks. Because variability can only be described in statistical terms, statistical methods play a central role in quality improvement; quality improvement is the reduction of variability in processes and products.

Chance causes and assignable causes

Basis	Chance (common) causes	Assignable (special) causes
Definition	Inherent or natural variability: the cumulative effect of many small, essentially unavoidable causes ("background noise").	Variability that is not part of the chance-cause pattern; generally large compared with the background noise.
Sources	Slight variations in material, machine vibration, ambient conditions, measurement error.	Improperly adjusted or controlled machines, operator errors, defective raw material.
Process state	Process operating with only chance causes is in statistical control (stable system of chance causes).	Process operating with assignable causes present is out of control.
Effect	Predictable within limits; part of the process.	Shifts the mean and/or increases the standard deviation; more output falls outside specifications.
Detection and removal	Cannot be removed without changing the process; requires management action.	Detected quickly by control charts; removed by investigation and corrective action at the process.



(a) Only chance causes: process in control

(b) Assignable cause 1: mean shifts up

(c) Assignable cause 2: mean shifts down and spread increases

(d) Assignable cause 3: spread increases

Figure 2: Chance and assignable causes of variation: in (a) only chance causes are present; in (b) to (d) assignable causes move the mean or increase the spread so that more output falls outside LSL and USL

A process operating with only chance causes of variation is **in statistical control**; a process operating in the presence of assignable causes is **out of control**. A control chart is used to detect assignable causes quickly, so that the root cause can be identified and corrective action taken before many nonconforming units are made.

Types of process variability

- **Stationary and uncorrelated:** the data vary around a fixed mean in a stable, predictable manner (the case for which Shewhart charts are designed).
- **Stationary and autocorrelated:** successive observations are dependent, with a tendency to move in long runs on either side of the mean.
- **Nonstationary:** the process drifts without any sense of a stable or fixed mean.

15	Explain any four statistical concepts used in control charts. Source: Unit I | SEE Feb/Mar 2026 (IM353IA), Q2b	08	2	3

(1) **Shewhart control chart model:** if w is a sample statistic with mean μ_w and standard deviation σ_w, then UCL $= \mu_w + L\sigma_w$, center line $= \mu_w$ and LCL $= \mu_w - L\sigma_w$, where L is the distance of the limits from the center line in standard deviation units. By the central limit theorem, a sample average is approximately normal with standard deviation $\sigma_{\bar{x}} = \sigma/\sqrt{n}$.

(2) **Three-sigma, probability and warning limits:** with $L = 3$ the probability of a type I error is 0.0027. In probability limits the type I error probability is chosen directly (for example, 0.001 gives 3.09-sigma limits). Warning limits are typically at two sigma.

(3) **Hypothesis testing and errors:** a point inside the limits corresponds to accepting that the process is in control and a point outside to rejecting it. Type I error is a false alarm (a point outside the limits with no assignable cause); type II error is a point inside the limits when the process is really out of control. Moving the limits farther out reduces type I and increases type II error.

(4) **Average run length (ARL):** the average number of points plotted before a point indicates an out-of-control condition, $ARL = 1/p$, where p is the probability that a point exceeds the limits; in control $ARL_0 = 1/0.0027 \approx 370$. The

average time to signal is $ATS = ARL \times h$, where h is the sampling interval in hours.

(5) **Rational subgroups:** samples are selected so that differences between subgroups are maximized when assignable causes are present, while differences within a subgroup are minimized.

(6) **Sample size and sampling frequency:** chart design includes the choice of sample size, control limits and sampling frequency.

Illustration: flow width of resist (process mean 1.5 micron, σ = 0.15 micron, n = 5)

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.15}{\sqrt{5}} = 0.0671$$

$$\text{UCL} = 1.5 + 3(0.0671) = 1.7013, \qquad \text{CL} = 1.5, \qquad \text{LCL} = 1.5 - 3(0.0671) = 1.2987$$

Warning limits: $1.5 \pm 2(0.0671) = 1.6342$ and 1.3658

16	Define and explain the type I and type II errors in the context of control charts. Are they related? How does the choice of control limits influence these two errors? Source: Unit I | SEE Fast Track Jul 2019 (16IM53), Q2c	06	–	–

Type I error (α): concluding that the process is out of control when it is really in control: a point falls outside the control limits although no assignable cause is present (a false alarm). With three-sigma limits and a normal statistic, $\alpha = 0.0027$.

Type II error (β): concluding that the process is in control when it is really out of control: a point falls inside the limits although the process has shifted. β is read from the operating-characteristic (OC) curve of the chart and depends on the size of the shift and the sample size.

Are they related? Effect of the control limits

Yes. For a given sample size the two risks move in opposite directions. Moving the control limits **farther** from the center line decreases the risk of type I error but increases the risk of type II error. Moving them **closer** increases the type I error and decreases the type II error. Three-sigma limits are a compromise that works well in practice; warning limits at two-sigma increase sensitivity (smaller β) at the cost of more false alarms. Increasing the sample size reduces β without increasing α.

Probability limits, in which the type I error probability is chosen directly (0.001 gives 3.09-sigma limits), are another way of setting this trade-off.

17	Explain the concept of Rational Subgrouping and its principles. Apply these principles to real-world scenarios in manufacturing or service industries to show how subgrouping separates common and special causes of variation and improves process stability. Source: Unit I | SEE Feb/Mar 2025 (IM353IA), Q2a; CIE-I, 26/11/2024; CIE-I, 07/11/2025; SEE Nov/Dec 2019 (16IM53), Q3b; SEE Jan/Feb 2021 (18IM53), Q4c (any four principles)	10	2	1

Rational subgroup concept (Shewhart): subgroups or samples should be selected so that if assignable causes are present, the chance for differences **between** subgroups is maximized, while the chance for differences due to assignable causes **within** a subgroup is minimized. The within-subgroup variation then represents only common (chance) causes and is used to set the control limits, so special causes show up as points out of control.

Principles of rational subgrouping

(1) **Use time order of production** as the logical basis for forming subgroups, because it allows assignable causes that occur over time to be detected.

(2) **Snapshot approach:** each subgroup consists of units produced at the same time or consecutively. It minimizes within-sample variation, maximizes between-sample variation when assignable causes are present, gives the best estimate of the process standard deviation and is used when the main purpose is to detect process shifts.

(3) **Random-sample approach:** each subgroup is a random sample of all output produced since the last sample. It is used when the chart supports decisions on accepting all units produced in the interval; care is needed because drift during the interval inflates the range and widens the limits.

(4) **Do not mix sources:** do not combine output of different machines, spindles, heads, operators or shifts in one subgroup; chart each source separately or use a group chart.

(5) **Do not spread a subgroup across a known change** (for example, the end of one shift and the start of the next), or differences between shifts will be hidden.
(6) **Choose subgroup size and frequency** to detect the shift of interest: larger samples detect small shifts; small, frequent samples are favoured in high-volume production.

Care must be taken in choosing the interval between samples, because any process can be made to appear in statistical control just by stretching out the interval between observations.

Application to real-world scenarios

- **Manufacturing (machining of shafts on a CNC lathe):** take 5 consecutive shafts every hour from each machine (snapshot). Within-subgroup variation reflects only machine noise; a tool change, wrong offset or new bar stock appears as a shift between subgroups on the $\bar{x}$ chart. Shafts from two lathes are never mixed in one subgroup.
- **Filling line with multiple heads:** form subgroups from one filling head at a time so that a blocked nozzle on one head is not hidden by the others.
- **Service (bank loan processing):** measure the processing time of 4 consecutive applications handled by the same officer in the same hour. Differences between hours, officers or days (special causes such as system downtime or staff shortage) then appear between subgroups.
- **Hospital emergency room:** subgroups of waiting times by shift separate common variation from special causes such as a night-shift staffing gap.

Proper rational subgrouping makes the control chart sensitive to special causes, prevents false alarms, gives correct estimates of process variability and so improves process stability.

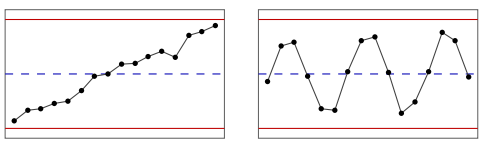
18	Explain the need for the change in the control limits of the control charts. Source: Unit I | SEE Jan/Feb 2021 (18IM53), Q2b	06	–	–

Control limits are not fixed for ever. Control charts are applied in two phases, and the limits are changed in each of them.

(1) **Phase I (trial limits):** a retrospective analysis of process data (20 to 25 subgroups) is used to construct trial control limits. It is effective in detecting large, sustained shifts in process parameters, outliers, measurement errors and data entry errors, and it helps to identify and remove assignable causes. Points outside the limits for which assignable causes are found are discarded and the limits are recalculated.
(2) **Phase II (monitoring):** the process is assumed to be reasonably stable and the emphasis is on monitoring, not on bringing an unruly process into control. The limits revised at the end of phase I are used for future production.
(3) **Process improvement or change:** when an assignable cause is permanently removed, or the process is changed (new machine, method, material, tooling, operator or product design), the mean or variability changes and new limits are computed from new data.
(4) **Periodic review:** limits are reviewed regularly (for example every 25, 50 or 100 samples) to check that they still represent the process.
(5) **Change in sample size or standards:** when the subgroup size or the target values change, the limits must be recomputed.

Limits should not be changed merely because points fall outside them; every revision should be based on a real, verified change in the process.

19	Identify and explain the causes of (i) trend and (ii) cyclic pattern in control charts. Source: Unit I | SEE Jan/Feb 2021 (18IM53), Q3a	06	–	–



(a) Trend (gradual drift) (b) Cycles

Figure 3: Trend and cyclic patterns on an $\bar{x}$ chart

(i) Trend: a continuous movement of the points in one direction (a run up or run down), also called drift. Causes: gradual wearing out or deterioration of a tool or another critical component, operator fatigue, gradual change in environmental

conditions (temperature, humidity), accumulation of waste products, seasonal effects. On the R chart a trend may come from gradual improvement in operator skill (downward) or wear (upward).

(ii) Cyclic pattern: a short trend in the data that occurs in repeated patterns. On the $\bar{x}$ chart it may be caused by systematic environmental changes (temperature), operator fatigue, regular rotation of operators or machines, fluctuation in voltage or pressure, or raw-material deliveries. On the R chart cycles may come from maintenance schedules or operator fatigue. Elimination of the cyclic source reduces variability and increases yield.

Recognising them on the chart

- A trend is a run up or a run down: for example, after the 4th point five points in a row increase in magnitude, and there is an unusually long run down beginning with the 18th point.
- A pattern is nonrandom when, for example, 19 of 25 points plot below the center line while only 6 plot above it.
- Sensitizing rules detect these patterns: six points in a row steadily increasing or decreasing (trend), fifteen points in a row in zone C, fourteen points in a row alternating up and down, and an unusual or nonrandom pattern in the data.
- When the cyclic pattern is eliminated the variability of the process is reduced and the process fits better inside the specification limits.

20	What are the two most common types of control charts for variables? Briefly explain their purpose. Source: Unit II | SEE Feb/Mar 2025 (IM353IA), Q3a	06	2	1

The two most common control charts for variables are the $\bar{x}$–R pair and, as the alternative pair, the $\bar{x}$–s charts. Samples (subgroups) of n units are measured.

$\bar{x}$ **chart:** plots the subgroup average and detects shifts in the process mean (location). Limits: $\bar{\bar{x}} \pm A_2\bar{R}$ (or $\bar{\bar{x}} \pm A_3\bar{s}$ when paired with the s chart).

R **chart:** plots the subgroup range (maximum minus minimum) and detects changes in the process variability (spread). Limits: $D_3\bar{R}$ and $D_4\bar{R}$; for n ≤ 6, $D_3 = 0$, so the LCL is not used.

s **chart:** plots the subgroup standard deviation, a statistically more efficient estimator of variability. Limits: $B_3\bar{s}$ and $B_4\bar{s}$.

Why chart both? The mean and the variability can shift independently: a process may stay centred while becoming more erratic, or drift off target while its spread stays constant. Both charts must be in control before the process is judged in control, and the R chart is examined first because the $\bar{x}$ limits depend on the process being stable in variability.

Aspect	**$\bar{x}$–R chart**	**$\bar{x}$–s chart**
Subgroup size	Small n (≤ 10), typically 4 to 6	Larger n (> 10)
Efficiency	Range is a less efficient estimator of σ as n grows	Standard deviation is a more efficient, more precise estimator of σ
Calculation	Very simple (max − min); ideal for manual shop-floor charting	Needs a standard deviation; suited to software or automated data collection
Sensitivity	Slightly less sensitive to changes in variability for large n	More sensitive; preferred for large subgroups

21	Explain the steps in constructing $\bar{X}$–R charts and interpret $\bar{X}$–R charts. Source: Unit II | SEE Feb/Mar 2026 (IM353IA), Q4a	10	3	2

Steps in constructing the $\bar{x}$–R charts

(1) Select the quality characteristic, the subgroup size (usually n = 4 or 5) and the sampling frequency using rational subgroups.
(2) Collect m = 20 to 25 subgroups of data in time order.
(3) For each subgroup compute the mean $\bar{x}_i$ and the range $R_i = x_{max} - x_{min}$.
(4) Compute the grand average $\bar{\bar{x}} = \sum \bar{x}_i/m$ and the average range $\bar{R} = \sum R_i/m$.
(5) Compute trial control limits. R chart: UCL $= D_4\bar{R}$, CL $= \bar{R}$, LCL $= D_3\bar{R}$. $\bar{x}$ chart: UCL $= \bar{\bar{x}} + A_2\bar{R}$, CL $= \bar{\bar{x}}$, LCL $= \bar{\bar{x}} - A_2\bar{R}$, with A_2, D_3, D_4 from the table of factors for the subgroup size.
(6) Plot the R chart first, then the $\bar{x}$ chart, with the subgroup numbers on the horizontal axis.
(7) Examine the charts for points outside the limits and nonrandom patterns. For points with identified assignable causes,

discard them and revise the limits (phase I).

(8) Use the revised limits for future monitoring (phase II) and estimate the process standard deviation $\hat{\sigma} = \bar{R}/d_2$ for capability analysis.

Interpretation of the $\bar{x}$–R charts

- Interpret the R chart first: if the variability is out of control, the $\bar{x}$ limits are not reliable.
- A process is in control when all the points are inside the limits and the pattern is random; a point outside the limits is evidence that it is out of control.
- **Western Electric rules:** (1) one or more points outside the control limits; (2) two of three consecutive points beyond the two-sigma warning limits; (3) four of five consecutive points at a distance of one sigma or more from the center line; (4) eight consecutive points on one side of the center line.
- **Additional sensitizing rules:** six points in a row steadily increasing or decreasing; fifteen points in a row in zone C; fourteen points in a row alternating up and down; eight points in a row on both sides of the center line with none in zone C; an unusual or nonrandom pattern; one or more points near a warning or control limit.
- Patterns and typical causes: cycles (environmental changes, operator rotation), trends (tool wear, fatigue), sudden shifts in level (new material, operator or setting), mixtures (two populations, over-control), stratification (points hugging the center line: wrong subgrouping or wrong limits).
- A change on the $\bar{x}$ chart indicates a change in the process mean; a change on the R chart indicates a change in uniformity (dispersion).

$$\alpha = 1 - \prod_{i=1}^{k}(1 - \alpha_i)$$

When k decision rules are used together and are independent, the overall type I error probability is α as given above (α_i is the type I error probability under rule i), so every additional rule makes the chart more sensitive but also raises the false alarm rate.

22	A manufacturing company measures the diameter (in mm) of a component during production. Data are collected in 6 samples of 4 measurements:	10	4	4

Sample	Meas. 1	Meas. 2	Meas. 3	Meas. 4
1	24.8	25.1	24.9	25.0
2	25.3	25.2	25.4	25.1
3	24.7	24.9	24.8	24.6
4	25.5	25.4	25.6	25.3
5	24.9	25.0	25.1	24.8
6	25.2	25.3	25.1	25.0

(a) Calculate the sample means. (b) Determine the $\bar{X}$-chart and R-chart control limits. (c) Interpret the control charts: are the process means and ranges in control? Identify any points or patterns that indicate special-cause variation.

Source: Unit II | CIE-I, 07/11/2025

(a) Sample means and ranges

Sample	x_1	x_2	x_3	x_4	$\bar{x}$	R
1	24.8	25.1	24.9	25.0	24.950	0.3
2	25.3	25.2	25.4	25.1	25.250	0.3
3	24.7	24.9	24.8	24.6	24.750	0.3
4	25.5	25.4	25.6	25.3	25.450	0.3
5	24.9	25.0	25.1	24.8	24.950	0.3
6	25.2	25.3	25.1	25.0	25.150	0.3

(b) Control limits

$$\bar{\bar{x}} = \frac{\sum \bar{x}_i}{m} = \frac{150.500}{6} = 25.0833, \qquad \bar{R} = \frac{\sum R_i}{m} = \frac{1.800}{6} = 0.3000$$

For $n = 4$: $A_2 = 0.729,\ D_3 = 0,\ D_4 = 2.282$

$\bar{x}$ chart: $\text{UCL} = \bar{\bar{x}} + A_2\bar{R} = 25.0833 + 0.729(0.3000) = 25.3020; \quad \text{CL} = 25.0833$

$$\text{LCL} = \bar{\bar{x}} - A_2\bar{R} = 25.0833 - 0.729(0.3000) = 24.8646$$

R chart: $\text{UCL} = D_4\bar{R} = 2.282(0.3000) = 0.6846; \quad \text{CL} = 0.3000; \quad \text{LCL} = D_3\bar{R} = 0.0000$

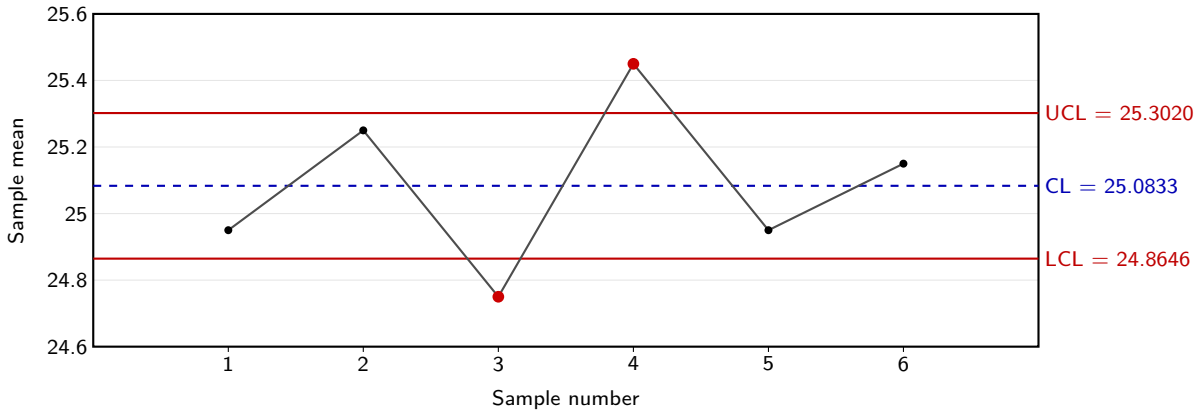


Figure 4: $\bar{x}$ chart

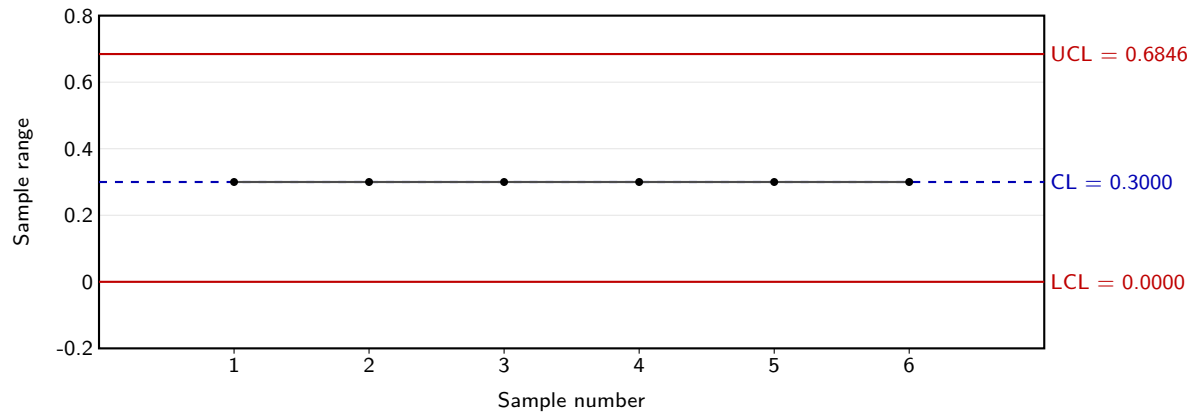


Figure 5: R chart

(c) Interpretation

All the sample ranges are equal (0.3 mm) and lie well inside the R-chart limits, so the process variability is in control. On the $\bar{x}$ chart, sample 3 and 4 fall outside the control limits (sample 3 = 24.750 is below the LCL of 24.8646, and sample 4 = 25.450 is above the UCL of 25.3020). The sample means also alternate between high and low values. The process mean is therefore **out of control**: special causes such as tool wear and re-setting, a change of material lot or operator adjustment are shifting the mean between samples. These causes must be found and removed and the limits recomputed (phase I) before the chart is used for monitoring.

Answer: $\bar{x}$ chart: UCL = 25.3020, CL = 25.0833, LCL = 24.8646 mm; R chart: UCL = 0.6846, CL = 0.3000, LCL = 0. Ranges are in control; the mean is out of control (samples 3, 4).

23 A manufacturing company measures the diameter of a component in millimetres during a production process. The data are collected in 5 samples, each with 4 measurements:

Sample	Meas. 1	Meas. 2	Meas. 3	Meas. 4
1	10.2	10.3	10.1	10.4
2	10.6	10.8	10.4	10.5
3	10.4	10.2	10.2	10.3
4	10.1	10.3	10.3	10.2
5	10.5	10.4	10.6	10.5

(i) Calculate the mean ($\bar{X}$) and range (R) for each sample. (ii) Determine control limits using the standard factors $A_2 = 0.729$, $D_3 = 0$, $D_4 = 2.282$. (iii) Interpret the control chart.

Source: Unit II | SEE Feb/Mar 2025 (IM353IA), Q4b | 10 | 3 | 3

(i) Mean and range of each sample

Sample	x_1	x_2	x_3	x_4	$\bar{x}$	R
1	10.2	10.3	10.1	10.4	10.250	0.3
2	10.6	10.8	10.4	10.5	10.575	0.4
3	10.4	10.2	10.2	10.3	10.275	0.2
4	10.1	10.3	10.3	10.2	10.225	0.2
5	10.5	10.4	10.6	10.5	10.500	0.2

(ii) Control limits

$$\bar{\bar{x}} = \frac{\sum \bar{x}_i}{m} = \frac{51.825}{5} = 10.3650, \qquad \bar{R} = \frac{\sum R_i}{m} = \frac{1.300}{5} = 0.2600$$

For $n = 4$: $A_2 = 0.729,\ D_3 = 0,\ D_4 = 2.282$

$\bar{x}$ chart: $\text{UCL} = \bar{\bar{x}} + A_2\bar{R} = 10.3650 + 0.729(0.2600) = 10.5545; \quad \text{CL} = 10.3650$

$$\text{LCL} = \bar{\bar{x}} - A_2\bar{R} = 10.3650 - 0.729(0.2600) = 10.1755$$

R chart: $\text{UCL} = D_4\bar{R} = 2.282(0.2600) = 0.5933; \quad \text{CL} = 0.2600; \quad \text{LCL} = D_3\bar{R} = 0.0000$

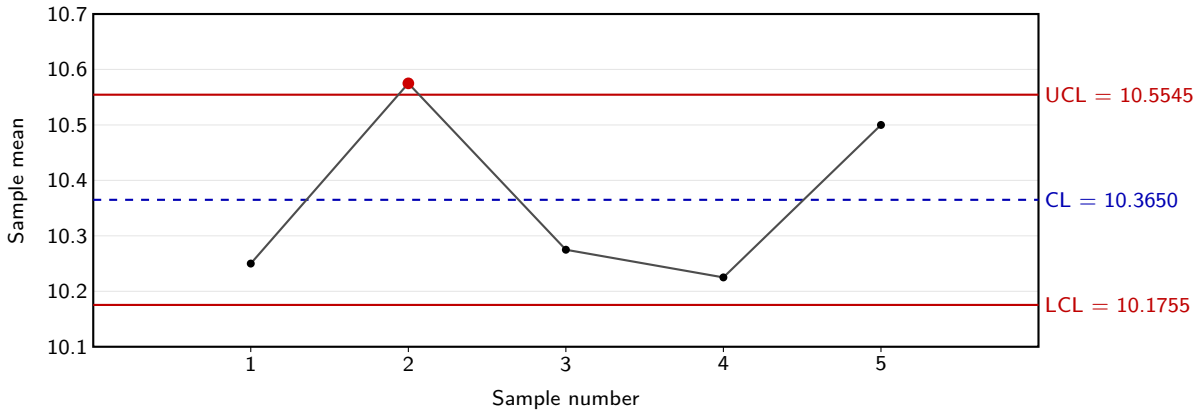


Figure 6: $\bar{x}$ chart

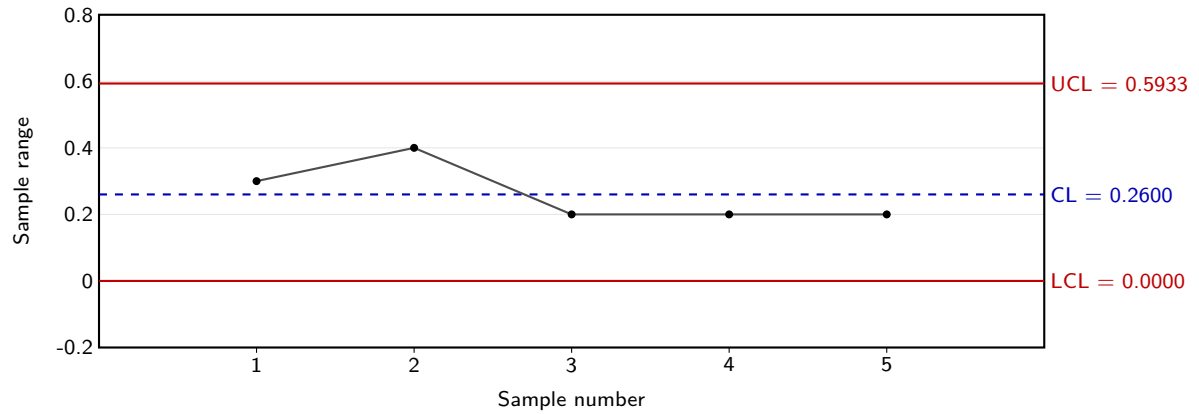


Figure 7: R chart

(iii) Interpretation

All ranges lie within the R-chart limits (0 to 0.593), so the process variability is in control. On the $\bar{x}$ chart, sample 2 ($\bar{x}$ = 10.575) lies above the UCL of 10.5545, so the process mean is **out of control**; sample 5 (10.500) is also close to the UCL. An assignable cause (for example, a machine setting or material change) affected sample 2 and should be investigated. After removing it, the limits should be recomputed from the remaining samples, and more samples (20 to 25) collected before the limits are used for routine control.

Answer: $\bar{x}$ chart: UCL = 10.5545, CL = 10.3650, LCL = 10.1755; R chart: UCL = 0.5933, CL = 0.2600, LCL = 0. Process mean is out of control (sample 2); variability is in control.

24 Mean values and ranges of data from 20 samples (sample size = 4) are shown below: | 10 | – | –

S.N.	Mean	Range	S.N.	Mean	Range
1	10	4	11	12	5
2	15	4	12	13	4
3	12	5	13	12	4
4	11	4	14	12	3
5	9	5	15	11	3
6	11	6	16	15	4
7	11	4	17	12	4
8	9	4	18	15	3
9	10	4	19	11	3
10	11	6	20	10	4

Calculate the control limits for $\bar{x}$ and R charts. Draw the control charts and give suitable conclusions.

Source: Unit II | SEE Jan/Feb 2021 (18IM53), Q3b

$$\bar{\bar{x}} = \frac{\sum \bar{x}_i}{m} = \frac{232.00}{20} = 11.600, \qquad \bar{R} = \frac{\sum R_i}{m} = \frac{83.00}{20} = 4.150$$

For $n = 4$: $A_2 = 0.729,\ D_3 = 0,\ D_4 = 2.282$

$\bar{x}$ chart: $\text{UCL} = \bar{\bar{x}} + A_2\bar{R} = 11.600 + 0.729(4.150) = 14.625; \quad \text{CL} = 11.600$

$$\text{LCL} = \bar{\bar{x}} - A_2\bar{R} = 11.600 - 0.729(4.150) = 8.575$$

R chart: $\text{UCL} = D_4\bar{R} = 2.282(4.150) = 9.470; \quad \text{CL} = 4.150; \quad \text{LCL} = D_3\bar{R} = 0.000$

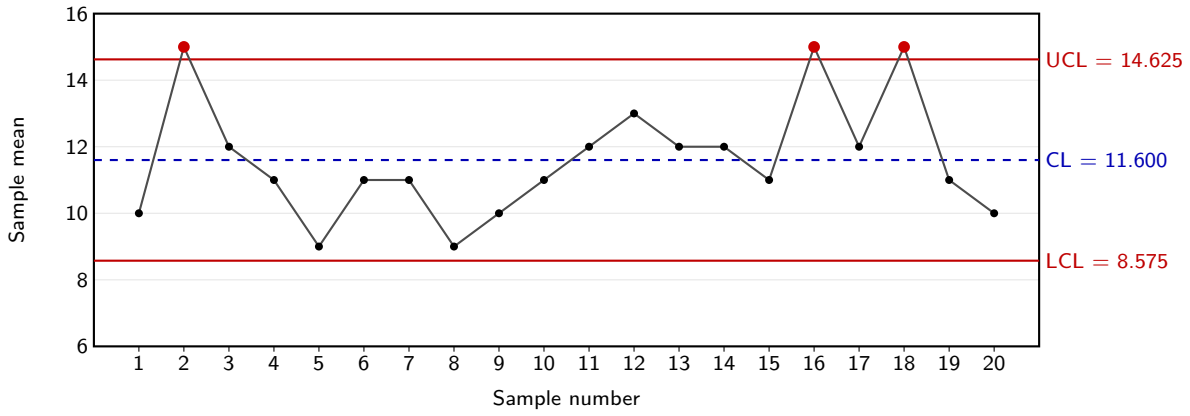


Figure 8: $\bar{x}$ chart

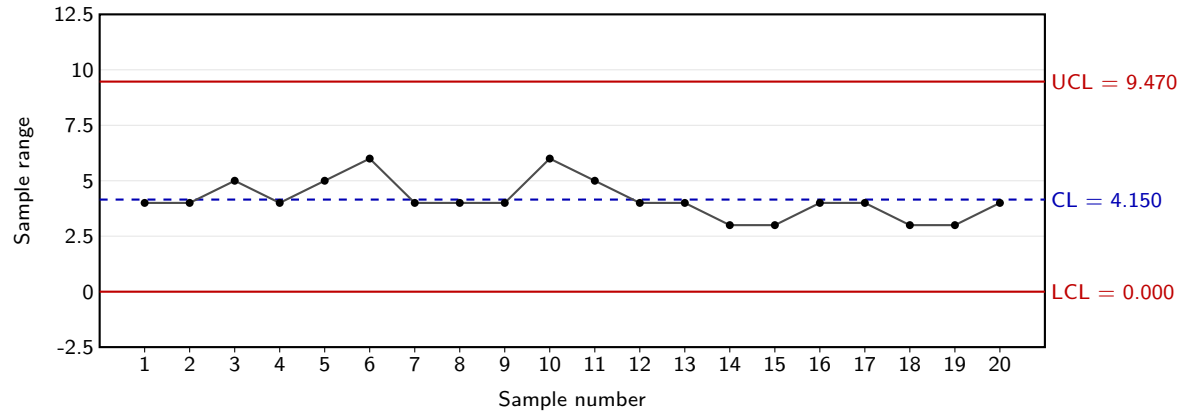


Figure 9: R chart

Conclusions

The R chart is in control: all ranges lie between 0 and 9.470. On the $\bar{x}$ chart, samples 2, 16, 18 ($\bar{x}$ = 15) are above the UCL of 14.625, and samples 5 and 8 ($\bar{x}$ = 9) are close to the LCL. The process variability is stable but the process mean is **not in statistical control**; assignable causes are shifting the mean upward from time to time.

Revised limits: if assignable causes are found for samples 2, 16, 18 and they are removed, the limits recomputed from the remaining 17 samples are $\bar{\bar{x}}$ = 11.000, $\bar{R}$ = 4.235, $\bar{x}$ chart UCL = 14.088 and LCL = 7.912, R chart UCL = 9.665. All remaining points lie within these revised limits.

Answer: Trial limits: $\bar{x}$ chart 14.625 / 11.600 / 8.575; R chart 9.470 / 4.150 / 0. Variability in control; mean out of control at samples 2, 16, 18.

25	In a foundry the specification on mould hardness is 65 ± 15. Moulds were chosen at random and mould hardness was estimated at 4 different places per mould. The data obtained are:	16	4	4

Mould No.	$\bar{X}$	R	Mould No.	$\bar{X}$	R
1	68	5	13	70	10
2	59	16	14	67	10
3	60	15	15	63	15
4	57	12	16	69	10
5	74	24	17	62	5
6	68	5	18	56	10
7	66	10	19	66	10
8	56	10	20	63	8
9	78	15	21	67	15
10	55	10	22	75	10
11	67	5	23	53	5
12	57	5	24	56	12

(i) Construct the control charts for the data. (ii) Interpret the state of the manufacturing process. (iii) Can the process meet the specification? (iv) What is the current state of affairs regarding non-conformity? (v) What are the different possible ways of reducing the rejection?

Source: Unit II | SEE Feb/Mar 2026 (IM353IA), Q3

(i) Control charts (n = 4, 24 moulds in the table)

$$\bar{\bar{x}} = \frac{\sum \bar{x}_i}{m} = \frac{1532.00}{24} = 63.833, \qquad \bar{R} = \frac{\sum R_i}{m} = \frac{252.00}{24} = 10.500$$

For $n = 4$: $A_2 = 0.729,\ D_3 = 0,\ D_4 = 2.282$

$\bar{x}$ chart: $\text{UCL} = \bar{\bar{x}} + A_2\bar{R} = 63.833 + 0.729(10.500) = 71.488; \quad \text{CL} = 63.833$

$$\text{LCL} = \bar{\bar{x}} - A_2\bar{R} = 63.833 - 0.729(10.500) = 56.179$$

R chart: $\text{UCL} = D_4\bar{R} = 2.282(10.500) = 23.961; \quad \text{CL} = 10.500; \quad \text{LCL} = D_3\bar{R} = 0.000$

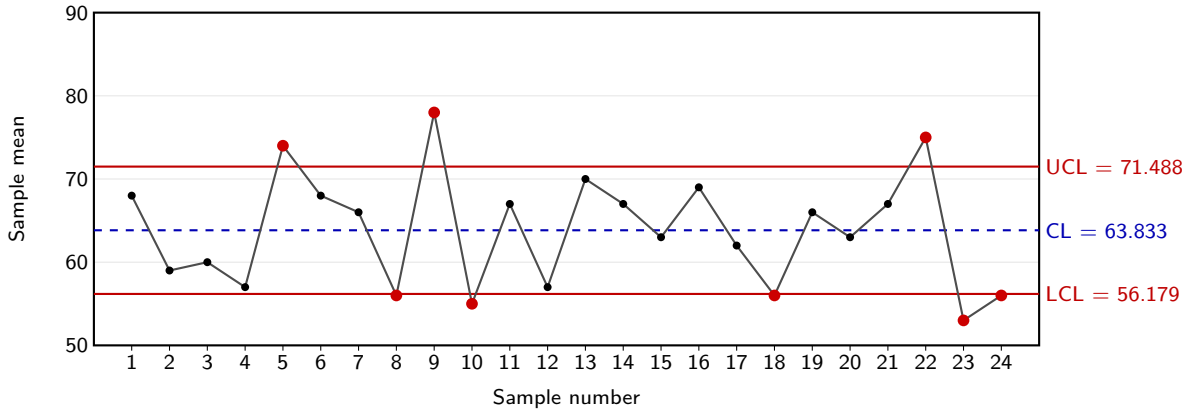


Figure 10: $\bar{x}$ chart

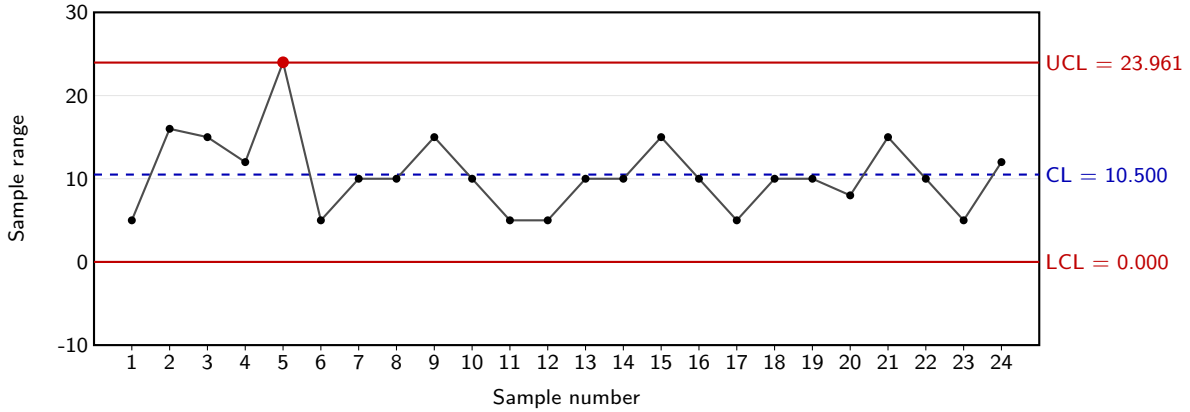


Figure 11: R chart

(ii) State of the process

R chart: mould 5 (R = 24) lies just above the UCL of 23.96; all other ranges are in control. The within-mould variability is therefore almost stable, with one assignable cause at mould 5. $\bar{x}$ **chart:** 8 of the 24 points lie outside the limits (moulds 5, 8, 9, 10, 18, 22, 23, 24); the hardness level swings widely from mould to mould. The process is **out of statistical control**, mainly in its mean; assignable causes (sand properties, pouring temperature, cooling time, metal composition) are present.

(iii) Can the process meet the specification of 65 ± 15?

$$\hat{\sigma} = \bar{R}/d_2, \quad 6\hat{\sigma} = 30.60, \quad \text{natural tolerance limits } \hat{\mu} \pm 3\hat{\sigma} = 48.53 \text{ to } 79.13$$

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{80.0 - 50.0}{30.60} = 0.980, \qquad C_{pk} = \frac{\min(USL - \hat{\mu},\ \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.904$$

$$P(x < LSL) = \Phi\left(\frac{50.0 - 63.83}{5.10}\right) = \Phi(-2.71) = 0.00334$$

$$P(x > USL) = 1 - \Phi\left(\frac{80.0 - 63.83}{5.10}\right) = 1 - \Phi(3.17) = 0.00076$$

$$\text{Total fraction nonconforming} = 0.00410\ (0.41\%)$$

No. Even on the basis of within-mould variation alone, the natural spread ($6\sigma = 30.60$) is slightly larger than the tolerance (30), so $C_p = 0.98 < 1$. Because the mean is also out of control, the actual spread of hardness is much wider than 6σ (the mould averages alone range from 53 to 78). The process cannot meet the specification in its present state.

(iv) Present state regarding non-conformity

Even if the mean were held at 63.83, about 0.41% of the hardness readings would fall outside 50 to 80 because of within-mould variation alone. Moulds whose averages are near or beyond the specification limits (moulds 5, 9, 22 and 23) are very likely to contain nonconforming hardness values, so the actual rejection is considerably higher than this estimate.

(v) Ways of reducing the rejection

- Identify and remove the assignable causes of mould-to-mould variation (control sand moisture, compactability, metal composition, pouring temperature and cooling time); use the $\bar{x}$ chart on-line.

- Keep the process mean centered at the nominal 65 so that both tails are equally far from the limits.
- Reduce within-mould variation by uniform ramming and standardized operating procedures; train operators.
- Use designed experiments to find the process settings that minimize hardness variability.
- If the process cannot be improved, review the specification with design or use a more capable process.

26	The following are $\bar{X}$ and R values of 20 subgroups of 5 readings:	16	–	–

Subgroup	$\bar{X}$	R	Subgroup	$\bar{X}$	R
1	34.0	4	11	38.4	4
2	31.6	2	12	34.0	14
3	30.8	3	13	35.0	4
4	33.8	5	14	33.8	7
5	31.6	3	15	31.6	5
6	33.0	5	16	33.0	7
7	28.2	13	17	32.6	3
8	33.8	19	18	31.8	9
9	37.8	6	19	35.6	6
10	35.8	4	20	33.0	4

(i) Determine the control limits of the $\bar{X}$ and R charts. (ii) What is the process capability? (iii) Is the process capable of meeting the specification limits? (iv) Determine the percentage of rejection, if any (specification limits are 33 ± 5). (v) What values of $\bar{X}$ and R do you recommend for future use?

Source: Unit II | SEE Jan/Feb 2021 (16IM53), Q4a

(i) Trial control limits

$$\bar{\bar{x}} = \frac{\sum \bar{x}_i}{m} = \frac{669.20}{20} = 33.460, \qquad \bar{R} = \frac{\sum R_i}{m} = \frac{127.00}{20} = 6.350$$

For $n = 5$: $A_2 = 0.577,\ D_3 = 0,\ D_4 = 2.114$

$\bar{x}$ chart: $\text{UCL} = \bar{\bar{x}} + A_2\bar{R} = 33.460 + 0.577(6.350) = 37.124; \quad \text{CL} = 33.460$

$$\text{LCL} = \bar{\bar{x}} - A_2\bar{R} = 33.460 - 0.577(6.350) = 29.796$$

R chart: $\text{UCL} = D_4\bar{R} = 2.114(6.350) = 13.424; \quad \text{CL} = 6.350; \quad \text{LCL} = D_3\bar{R} = 0.000$

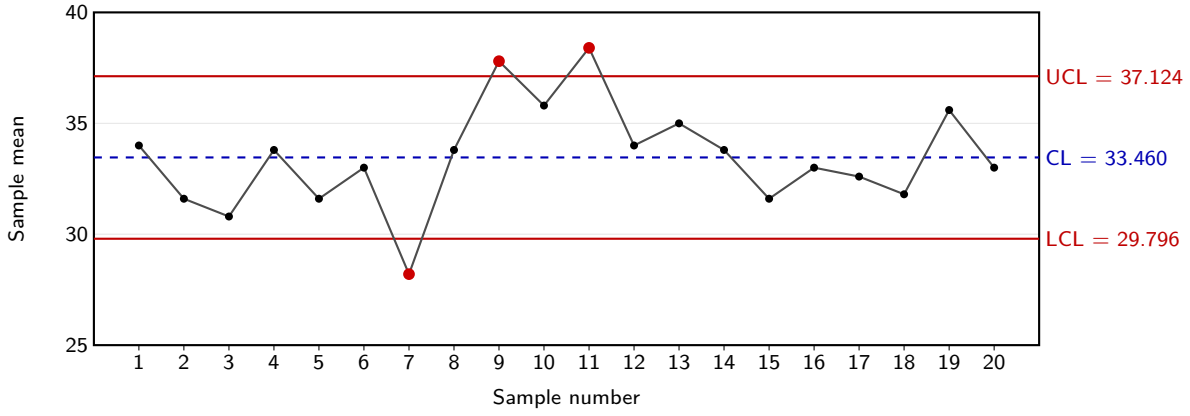


Figure 12: $\bar{x}$ chart

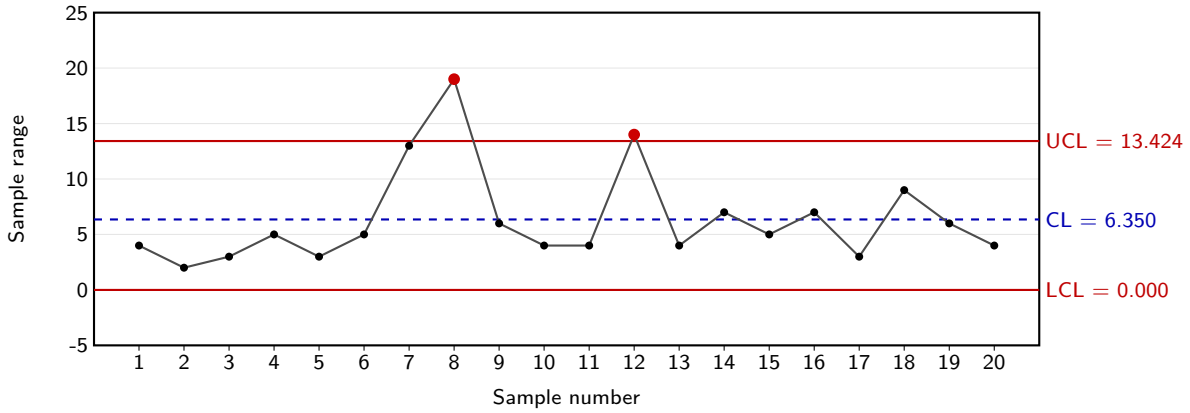


Figure 13: R chart

Out-of-control points: $\bar{x}$ chart subgroups 7, 9, 11; R chart subgroups 8, 12. Assuming assignable causes are found for them, these subgroups are discarded and the limits revised until all remaining points are in control (subgroups removed: 7, 8, 9, 11, 12).

Revised (15 subgroups): $\bar{\bar{x}} = 33.133,\ \bar{R} = 4.733$

$\bar{x}$ chart: UCL $= 35.864,$ LCL $= 30.402;$ R chart: UCL $= 10.006,$ LCL $= 0$

(ii)–(iv) Process capability and rejection (specification 33 ± 5)

$$\hat{\sigma} = \bar{R}/d_2, \quad 6\hat{\sigma} = 12.21, \quad \text{natural tolerance limits } \hat{\mu} \pm 3\hat{\sigma} = 27.03 \text{ to } 39.24$$

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{38.0 - 28.0}{12.21} = 0.819, \qquad C_{pk} = \frac{\min(USL - \hat{\mu},\ \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.797$$

$$P(x < LSL) = \Phi\left(\frac{28.0 - 33.13}{2.03}\right) = \Phi(-2.52) = 0.00583$$

$$P(x > USL) = 1 - \Phi\left(\frac{38.0 - 33.13}{2.03}\right) = 1 - \Phi(2.39) = 0.00839$$

Total fraction nonconforming $= 0.01421$ (1.42%)

Process capability = 6σ = 12.210, which is greater than the tolerance of 10, so C_p = 0.82: the process is **not capable**. The mean (33.133) is slightly above the nominal 33, giving C_{pk} = 0.80. Even after the assignable causes are removed, about 1.42% of the output is expected to fall outside 28 to 38, so the process cannot reliably meet the specification; the process variability must be reduced (or the tolerance reviewed) and the mean centred at 33.

(v) Values recommended for future use

Answer: Use $\bar{\bar{x}}$ = 33.133 and $\bar{R}$ = 4.733 (revised): $\bar{x}$ chart limits 30.402 to 35.864, R chart UCL 10.006. Process capability 6σ = 12.210 (C_p = 0.82), with about 1.42% rejection expected.

27	Samples of size $n = 5$ are taken from a manufacturing process every hour. A quality characteristic is measured and $\bar{x}$ and R are computed for each sample. After 25 samples have been analyzed, $\sum \bar{x}_i = 662.50$ and $\sum R_i = 9.00$. The quality characteristic is normally distributed. (i) Find the control limits for the $\bar{x}$ and R charts. (ii) Assume that both charts exhibit control. If the specifications are 26.40 ± 0.50, estimate the fraction nonconforming. (iii) If the mean of the process were 26.40, what fraction nonconforming would result? Source: Unit II | SEE Fast Track Oct 2020 (16IM53), Q3a	08	–	–

(i) Control limits

$$\bar{\bar{x}} = \frac{662.50}{25} = 26.5000, \qquad \bar{R} = \frac{9.00}{25} = 0.3600$$

$(n = 5:\ A_2 = 0.577,\ D_3 = 0,\ D_4 = 2.114,\ d_2 = 2.326)$

$\bar{x}$ chart: $\text{UCL} = 26.5000 + 0.577(0.3600) = 26.7077, \quad \text{CL} = 26.5000$

$$\text{LCL} = 26.5000 - 0.577(0.3600) = 26.2923$$

R chart: $\text{UCL} = 2.114(0.3600) = 0.7610, \quad \text{CL} = 0.3600, \quad \text{LCL} = 0(0.3600) = 0.0000$

$\hat{\sigma} = \bar{R}/d_2 = 0.3600/2.326 = 0.15477$

(ii) Fraction nonconforming (specifications 25.90 to 26.90)

$\hat{\sigma} = \bar{R}/d_2, \quad 6\hat{\sigma} = 0.929, \quad$ natural tolerance limits $\hat{\mu} \pm 3\hat{\sigma} = 26.036$ to 26.964

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{26.90 - 25.90}{0.929} = 1.077, \qquad C_{pk} = \frac{\min(USL - \hat{\mu},\ \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.861$$

$$P(x < LSL) = \Phi\left(\frac{25.90 - 26.500}{0.155}\right) = \Phi(-3.88) = 0.00005$$

$$P(x > USL) = 1 - \Phi\left(\frac{26.90 - 26.500}{0.155}\right) = 1 - \Phi(2.58) = 0.00488$$

Total fraction nonconforming $= 0.00493$ (0.49%)

(iii) If the process were centered at 26.40

$$P = 2\,\Phi\left(\frac{25.90 - 26.40}{0.1548}\right) = 2\,\Phi(-3.23) = 0.00124$$

Answer: (i) $\bar{x}$ chart 26.7077 / 26.500 / 26.2923; R chart 0.7610 / 0.360 / 0. (ii) Fraction nonconforming = 0.00493 (0.49%). (iii) With the mean at 26.40 it falls to 0.00124 (0.124%).

28	Samples of $n = 5$ units are taken from a process every hour. The $\bar{x}$ and R values for a particular quality characteristic are determined. After 25 samples have been collected, $\bar{\bar{x}} = 20$ and $\bar{R} = 4.56$. (i) What are the three-sigma control limits? (ii) Both charts exhibit control; estimate the process standard deviation. (iii) Assuming that the process output is normally distributed, if the specifications are 19 ± 5, what are your conclusions regarding the process capability? (iv) If the process mean shifts to 24, what is the probability of not detecting this shift on the first subsequent sample? Source: Unit II | SEE Fast Track Oct 2020 (16IM53), Q3b	08	–	–

(i) Three-sigma control limits and (ii) process standard deviation

$$\bar{\bar{x}} = 20,\ \bar{R} = 4.56,\ n = 5:\ A_2 = 0.577,\ D_3 = 0,\ D_4 = 2.114,\ d_2 = 2.326$$

$\bar{x}$ chart: $\text{UCL} = 20 + 0.577(4.56) = 22.6311, \quad \text{CL} = 20, \quad \text{LCL} = 20 - 0.577(4.56) = 17.3689$

R chart: $\text{UCL} = 2.114(4.56) = 9.6398, \quad \text{CL} = 4.56, \quad \text{LCL} = 0$

$$\hat{\sigma} = \bar{R}/d_2 = 4.56/2.326 = 1.9604$$

(iii) Process capability (specifications $19 \pm 5 = 14$ to 24)

$$\hat{\sigma} = \bar{R}/d_2, \quad 6\hat{\sigma} = 11.76, \quad \text{natural tolerance limits } \hat{\mu} \pm 3\hat{\sigma} = 14.12 \text{ to } 25.88$$

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{24.0 - 14.0}{11.76} = 0.850, \qquad C_{pk} = \frac{\min(USL - \hat{\mu},\ \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.680$$

$$P(x < LSL) = \Phi\left(\frac{14.0 - 20.00}{1.96}\right) = \Phi(-3.06) = 0.00110$$

$$P(x > USL) = 1 - \Phi\left(\frac{24.0 - 20.00}{1.96}\right) = 1 - \Phi(2.04) = 0.02066$$

Total fraction nonconforming $= 0.02176$ (2.18%)

$C_p = 0.850 < 1$: the natural spread (11.763) is larger than the tolerance (10), and the mean (20) is 1 unit above the nominal 19. The process is **not capable**; about 2.18% of the output is nonconforming. Variability must be reduced and the mean re-centred at 19.

(iv) Probability of not detecting a shift to 24 on the first sample

$$\sigma_{\bar{x}} = \hat{\sigma}/\sqrt{n} = 1.9604/\sqrt{5} = 0.8767$$

$$\beta = P(\text{LCL} \le \bar{x} \le \text{UCL} \mid \mu = 24) = \Phi\left(\frac{22.6311 - 24}{0.8767}\right) - \Phi\left(\frac{17.3689 - 24}{0.8767}\right)$$

$$= \Phi(-1.56) - \Phi(-7.56) = 0.0592$$

Answer: Limits: $\bar{x}$ 22.631 / 20 / 17.369, R 9.640 / 4.56 / 0; $\hat{\sigma}$ = 1.9604; C_p = 0.850, not capable (2.18% nonconforming); probability of not detecting the shift on the first sample = 0.0592.

29	Samples of $n = 6$ items each are taken from a process at regular intervals. A quality characteristic is measured and $\bar{X}$ and R are calculated for each sample. After 50 samples, $\sum \bar{X}_i = 2000$ and $\sum R_i = 200$. Assume that the quality characteristic is normally distributed. (i) Compute the control limits for the $\bar{X}$ and R charts. (ii) All points fall between the control limits. What are the natural tolerance limits of the process? (iii) If the specification limits are 41 ± 5.0, what are your conclusions regarding the ability of the process to produce items within these specifications? (iv) If an item exceeding the upper specification limit can be reworked and one below the lower specification limit must be scrapped, what percent of scrap and rework is the process producing? Source: Unit II | SEE Mar/Apr 2023 (18IM53), Q3a	12	–	–

(i) Control limits

$$\bar{\bar{x}} = \frac{2000.00}{50} = 40.0000, \qquad \bar{R} = \frac{200.00}{50} = 4.0000$$

$(n = 6:\ A_2 = 0.483,\ D_3 = 0,\ D_4 = 2.004,\ d_2 = 2.534)$

$\bar{x}$ chart: $\text{UCL} = 40.0000 + 0.483(4.0000) = 41.9320, \quad \text{CL} = 40.0000$

$$\text{LCL} = 40.0000 - 0.483(4.0000) = 38.0680$$

R chart: $\text{UCL} = 2.004(4.0000) = 8.0160, \quad \text{CL} = 4.0000, \quad \text{LCL} = 0(4.0000) = 0.0000$

$\hat{\sigma} = \bar{R}/d_2 = 4.0000/2.534 = 1.57853$

(ii) Natural tolerance limits

$$\hat{\mu} \pm 3\hat{\sigma} = 40 \pm 3(1.5785) = 40 \pm 4.736 \Rightarrow 35.264 \text{ to } 44.736$$

(iii) Ability to meet specifications 41 ± 5 (36 to 46)

$\hat{\sigma} = \bar{R}/d_2, \quad 6\hat{\sigma} = 9.47, \quad$ natural tolerance limits $\hat{\mu} \pm 3\hat{\sigma} = 35.26$ to 44.74

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{46.0 - 36.0}{9.47} = 1.056, \qquad C_{pk} = \frac{\min(USL - \hat{\mu},\ \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.845$$

$$P(x < LSL) = \Phi\left(\frac{36.0 - 40.00}{1.58}\right) = \Phi(-2.53) = 0.00564$$

$$P(x > USL) = 1 - \Phi\left(\frac{46.0 - 40.00}{1.58}\right) = 1 - \Phi(3.80) = 0.00007$$

Total fraction nonconforming $= 0.00571$ (0.57%)

The natural spread (9.471) is slightly smaller than the tolerance (10), so C_p = 1.056, just above 1 (marginally capable). However, the process mean (40) is 1 unit below the nominal 41, so C_{pk} = 0.845 < 1: the process is not centered and will produce some items below the LSL. Re-centering the mean at 41 would bring the process to its potential capability.

(iv) Scrap and rework

Answer: Scrap (below LSL 36) = 0.564%; rework (above USL 46) = 0.0072%; total = 0.571%.

30	Subgroups of 5 items each are taken from a manufacturing process at regular intervals. A certain quality characteristic is measured and $\bar{X}$ and R values computed for each subgroup. After 25 subgroups, $\sum \bar{X} = 357.5$ and $\sum R = 8.8$. Assume that all the points are within the control limits on both the charts. The specifications are 14.4 ± 0.4. (i) Compute the control limits for the $\bar{X}$ and R charts. (ii) What is the process capability? (iii) Determine the percentage of rejections, if any. (iv) What can you conclude regarding its ability to meet the specifications? Source: Unit II | SEE Nov/Dec 2019 (16IM53), Q4a	12	–	–

(i) Control limits

$$\bar{\bar{x}} = \frac{357.50}{25} = 14.3000, \qquad \bar{R} = \frac{8.80}{25} = 0.3520$$

$$(n = 5:\ A_2 = 0.577,\ D_3 = 0,\ D_4 = 2.114,\ d_2 = 2.326)$$

$\bar{x}$ chart: $\text{UCL} = 14.3000 + 0.577(0.3520) = 14.5031, \quad \text{CL} = 14.3000$

$$\text{LCL} = 14.3000 - 0.577(0.3520) = 14.0969$$

R chart: $\text{UCL} = 2.114(0.3520) = 0.7441, \quad \text{CL} = 0.3520, \quad \text{LCL} = 0(0.3520) = 0.0000$

$$\hat{\sigma} = \bar{R}/d_2 = 0.3520/2.326 = 0.15133$$

(ii) Process capability

Process capability $= 6\hat{\sigma} = 6(0.1513) = 0.9080$; the natural tolerance limits are 14.30 ± 0.4540, i.e. 13.846 to 14.754.

(iii) Percentage of rejections (specifications 14.0 to 14.8)

$\hat{\sigma} = \bar{R}/d_2, \quad 6\hat{\sigma} = 0.91, \quad$ natural tolerance limits $\hat{\mu} \pm 3\hat{\sigma} = 13.85$ to 14.75

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{14.8 - 14.0}{0.91} = 0.881, \qquad C_{pk} = \frac{\min(USL - \hat{\mu},\ \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.661$$

$$P(x < LSL) = \Phi\left(\frac{14.0 - 14.30}{0.15}\right) = \Phi(-1.98) = 0.02372$$

$$P(x > USL) = 1 - \Phi\left(\frac{14.8 - 14.30}{0.15}\right) = 1 - \Phi(3.30) = 0.00048$$

Total fraction nonconforming $= 0.02419$ (2.42%)

(iv) Conclusion

The process spread (0.9080) is greater than the tolerance (0.8), so C_p = 0.881 < 1, and the mean (14.3) is 0.1 below the nominal 14.4, giving C_{pk} = 0.661. The process is **not capable** of meeting the specifications; about 2.42% will be rejected, mostly below the LSL. The mean should be re-centred at 14.4 (which reduces rejections to about 0.82%) and the process variability must be reduced.

Answer: $\bar{x}$ chart 14.5031 / 14.300 / 14.0969; R chart 0.7441 / 0.352 / 0; capability $6\sigma = 0.9080$; rejection ≈ 2.42%; process not capable.

31	Subgroups of 4 items each are taken from a manufacturing process at regular intervals. A certain quality characteristic is measured and $\bar{X}$, R values are computed for each subgroup. After 25 subgroups, $\sum \bar{X} = 15350$ and $\sum R = 411.4$. (i) Compute the control limits for the $\bar{X}$–R charts. (ii) Assume all the points fall within the control limits on both charts. The specification limits are 610 ± 15. If the quality characteristic is normally distributed, what percentage of product would fail to meet specifications? (iii) Any product that falls below the LSL will be scrapped and above the USL must be reworked. It is suggested that the process be centered at a level so that not more than 0.1% of the product will be scrapped. What should be the aimed value of $\bar{X}'$ to make the scrap exactly 0.1%? Source: Unit II | SEE Fast Track Jul 2019 (16IM53), Q3a	12	–	–

(i) Control limits

$$\bar{\bar{x}} = \frac{15350.00}{25} = 614.000, \qquad \bar{R} = \frac{411.40}{25} = 16.456$$

$$(n = 4:\ A_2 = 0.729,\ D_3 = 0,\ D_4 = 2.282,\ d_2 = 2.059)$$

$\bar{x}$ chart: $\text{UCL} = 614.000 + 0.729(16.456) = 625.996, \quad \text{CL} = 614.000$

$$\text{LCL} = 614.000 - 0.729(16.456) = 602.004$$

R chart: $\text{UCL} = 2.282(16.456) = 37.553, \quad \text{CL} = 16.456, \quad \text{LCL} = 0(16.456) = 0.000$

$$\hat{\sigma} = \bar{R}/d_2 = 16.456/2.059 = 7.9922$$

(ii) Percentage failing the specifications 610 ± 15 (595 to 625)

$\hat{\sigma} = \bar{R}/d_2, \quad 6\hat{\sigma} = 48.0, \quad$ natural tolerance limits $\hat{\mu} \pm 3\hat{\sigma} = 590.0$ to 638.0

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{625 - 595}{48.0} = 0.626, \qquad C_{pk} = \frac{\min(USL - \hat{\mu},\ \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.459$$

$$P(x < LSL) = \Phi\left(\frac{595 - 614.0}{8.0}\right) = \Phi(-2.38) = 0.00872$$

$$P(x > USL) = 1 - \Phi\left(\frac{625 - 614.0}{8.0}\right) = 1 - \Phi(1.38) = 0.08436$$

Total fraction nonconforming $= 0.09308$ (9.31%)

(iii) Aimed value for exactly 0.1% scrap

$$P(x < 595) = 0.001 \Rightarrow \frac{595 - \bar{X}'}{\hat{\sigma}} = -3.09$$

$$\bar{X}' = 595 + 3.09(7.992) = 619.70$$

$$\text{Rework then} = 1 - \Phi\left(\frac{625 - 619.70}{7.992}\right) = 1 - \Phi(0.664) = 0.2535$$

Answer: Limits: $\bar{X}$ chart 626.00 / 614.0 / 602.00, R chart 37.55 / 16.456 / 0. Product failing specifications = 9.31% (scrap 0.87%, rework 8.44%). Aim $\bar{X}'$ = 619.70 for 0.1% scrap (rework then rises to about 25.3%).

32	A process is producing shafts with a nominal diameter of 50 mm. The specification limits are 50 ± 0.5 mm. From control chart data, the process mean is 50.1 mm and the process standard deviation (σ) is 0.1 mm. (a) Calculate the upper control limit (UCL) and lower control limit (LCL) for the process using $\pm 3\sigma$ limits. (b) Compare the control limits with the specification limits and comment on the process capability. Source: Unit II | CIE-I, 07/11/2025	10	4	4

(a) Three-sigma limits of the process

$$\text{UCL} = \mu + 3\sigma = 50.1 + 3(0.1) = 50.4 \text{ mm}$$

$$\text{CL} = 50.1 \text{ mm}$$

$$\text{LCL} = \mu - 3\sigma = 50.1 - 3(0.1) = 49.8 \text{ mm}$$

(b) Comparison with specification limits (49.5 to 50.5 mm)

$$C_p = \frac{USL - LSL}{6\sigma} = \frac{50.5 - 49.5}{6(0.1)} = 1.667$$

$$C_{pk} = \min\left(\frac{50.5 - 50.1}{0.3}, \frac{50.1 - 49.5}{0.3}\right) = \min(1.333,\ 2.000) = 1.333$$

$$P(x > 50.5) = 1 - \Phi(4.0) = 0.00003, \quad P(x < 49.5) = \Phi(-6.0) \approx 0$$

The natural tolerance limits (49.8 to 50.4 mm) lie inside the specification limits (49.5 to 50.5 mm), so the process is **capable**: C_p = 1.67 and C_{pk} = 1.33, both at or above the recommended 1.33, and practically no shafts (about 30 ppm) will be outside the specification. The mean is 0.1 mm above the nominal, so the upper side has less margin; re-centering at 50.0 mm would raise C_{pk} to 1.67. Note that control limits describe the process, while specification limits describe the requirement; they should not be confused.

Answer: UCL = 50.4 mm, LCL = 49.8 mm; process is capable (Cp = 1.67, Cpk = 1.33) but slightly off-centre.

33	A TiW layer is deposited on a substrate using a sputtering tool. The following table contains layer thickness measurements (in Angstroms) on 10 subgroups of four substrates. Set up $\bar{X}$ and s charts on this process and compute trial control limits.	10	3	CO3

Subgroup	x_1	x_2	x_3	x_4
1	459	449	435	450
2	443	440	442	442
3	457	444	449	444
4	469	463	453	438
5	443	457	445	454
6	444	456	456	457
7	445	449	450	445
8	446	455	449	452
9	444	452	457	440
10	432	463	463	443

Source: Unit II | CIE-II, 07/01/2025

For each subgroup $\bar{x}_i = \sum x/n$ and $s_i = \sqrt{\sum (x - \bar{x}_i)^2/(n-1)}$.

Subgroup	x_1	x_2	x_3	x_4	$\bar{x}$	s
1	459	449	435	450	448.25	9.912
2	443	440	442	442	441.75	1.258
3	457	444	449	444	448.50	6.137
4	469	463	453	438	455.75	13.549
5	443	457	445	454	449.75	6.801
6	444	456	456	457	453.25	6.185
7	445	449	450	445	447.25	2.630
8	446	455	449	452	450.50	3.873
9	444	452	457	440	448.25	7.676
10	432	463	463	443	450.25	15.392

$$\bar{\bar{x}} = \frac{4493.50}{10} = 449.350, \qquad \bar{s} = \frac{73.413}{10} = 7.341 \qquad (n = 4:\ A_3 = 1.628,\ B_3 = 0,\ B_4 = 2.266)$$

$\bar{x}$ chart: $\text{UCL} = \bar{\bar{x}} + A_3\bar{s} = 449.350 + 1.628(7.341) = 461.302;\quad \text{CL} = 449.350;\quad \text{LCL} = 437.398$

s chart: $\text{UCL} = B_4\bar{s} = 2.266(7.341) = 16.635;\quad \text{CL} = 7.341;\quad \text{LCL} = B_3\bar{s} = 0$

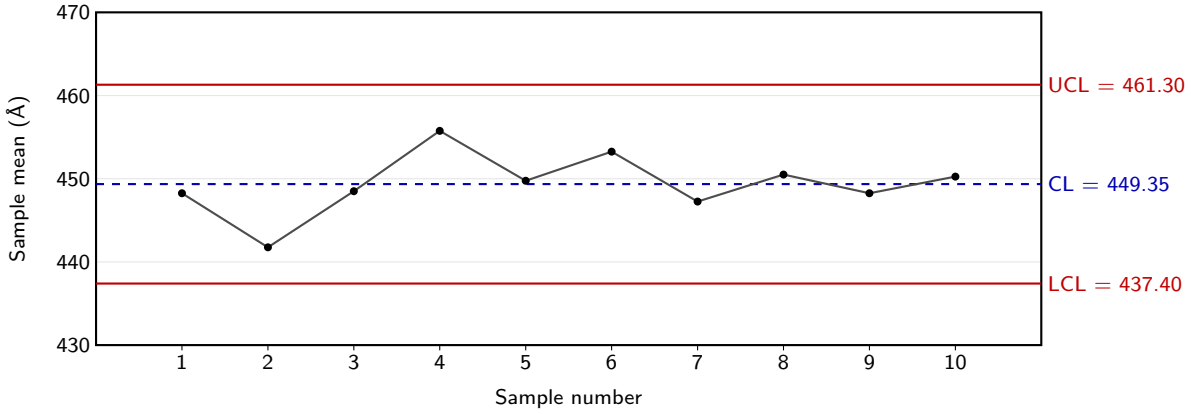


Figure 14: $\bar{x}$ chart for TiW layer thickness

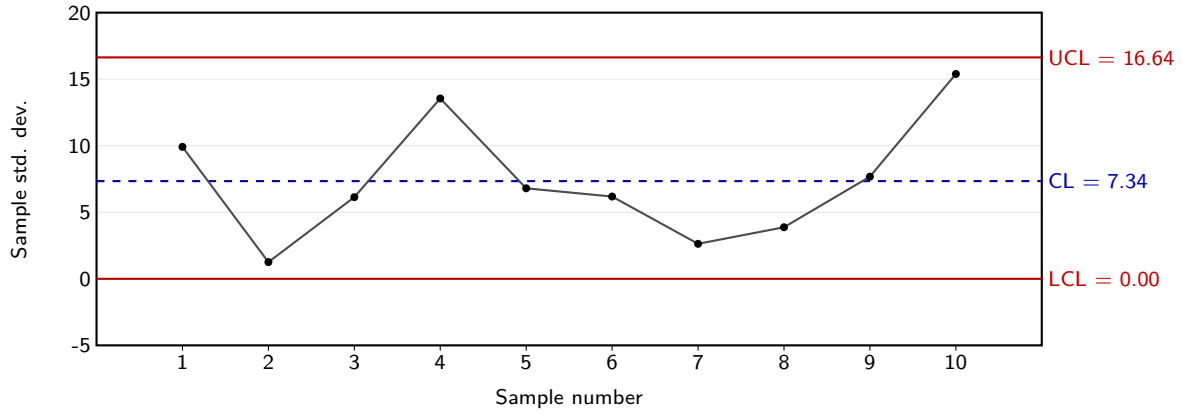


Figure 15: s chart for TiW layer thickness

All subgroup averages and standard deviations lie within the trial control limits and show no unusual pattern, so the sputtering process is in statistical control. The trial limits can be adopted for monitoring future production.

Answer: $\bar{x}$ chart: UCL = 461.30, CL = 449.35, LCL = 437.40 Å; s chart: UCL = 16.64, CL = 7.34, LCL = 0.

34	When do we use the moving range chart? Explain. Source: Unit II | SEE Nov/Dec 2019 (16IM53), Q4b	04	–	–

The moving-range chart is used together with the chart for individual measurements (X–MR chart) when only **one observation per sampling period** is available or practical, so that subgrouping is not possible and the variability cannot be estimated from a subgroup range. Typical situations:

- Destructive testing, where only one unit can be tested per batch.
- A very low production rate, for example one batch per day.
- Homogeneous or bulk processes, such as chemical batches and tank fills, where repeat measurements differ only by laboratory or analysis error.
- Automated 100% inspection, where every unit is measured individually.

The moving range is computed from each pair of consecutive observations, $MR_i = |x_i - x_{i-1}|$, and σ is estimated indirectly from it: $\hat{\sigma} = \overline{MR}/d_2 = \overline{MR}/1.128$. Control limits: X chart $\bar{x} \pm 2.66\,\overline{MR}$; MR chart UCL $= 3.267\,\overline{MR}$, LCL $= 0$ (the constants correspond to n = 2).

Points to note: consecutive MR values are correlated (each x_i appears in two MR terms), so the MR chart is interpreted with caution and mainly used to estimate σ; the X chart is the primary chart for detecting shifts; and the X–MR combination is less sensitive than $\bar{x}$–R at detecting small shifts.

35	Write a note on the standardized control chart. What is process capability? Explain any one of the techniques of process capability analysis. Source: Unit II | CIE-II, 07/01/2025	10	2	CO1

Standardized control chart

A standardized control chart plots the statistic in standard deviation units, so the control limits are always at ± 3 and the center line at zero. For a p chart with variable sample size the standardized value is $Z_i = (\hat{p}_i - \bar{p})/\sqrt{\bar{p}(1-\bar{p})/n_i}$; for an $\bar{x}$ chart $Z_i = (\bar{x}_i - \mu)/(\sigma/\sqrt{n})$. It is useful when the sample size varies (limits do not change from sample to sample), in short production runs where different part numbers are plotted on one chart (deviation from nominal), and it makes the runs and pattern rules easy to apply.

Process capability

Process capability describes the natural, inherent variability of a process once it is in a state of statistical control, that is, once only chance causes remain. It answers the question whether the process can consistently meet the specifications. It is measured by the natural tolerance 6σ (limits $\mu \pm 3\sigma$) and compared with the specification limits through $C_p = (USL - LSL)/6\sigma$ and C_{pk}.

Methods of estimating process capability

- **Histogram or probability plot:** plot the measurements of a large sample and compare the spread and shape with the specification limits; simple and intuitive, but it does not confirm that the process is in statistical control.
- **Control chart method:** run an $\bar{x}$–R (or $\bar{x}$–s) chart until the process shows statistical control, then estimate $\hat{\sigma} = \bar{R}/d_2$ (or $\bar{s}/c_4$). This is the preferred method because it confirms stability before capability is estimated.
- **Designed experiments:** study how process variables affect the output characteristic so that capability can be predicted and optimized at the design stage; most rigorous but resource-intensive.

Technique: capability analysis using a control chart

(1) Collect 20 to 25 subgroups and construct $\bar{x}$ and R charts; bring the process into statistical control by removing assignable causes.
(2) Estimate the process mean $\hat{\mu} = \bar{\bar{x}}$ and standard deviation $\hat{\sigma} = \bar{R}/d_2$.
(3) Compute the natural tolerance limits $\hat{\mu} \pm 3\hat{\sigma}$ and the capability ratios C_p and C_{pk}.
(4) Estimate the fraction nonconforming from the normal distribution: $P(x < LSL) + P(x > USL)$.

36	What is process capability? Explain the procedure of computing process capability using the average range method. Source: Unit II | SEE Fast Track Jul 2019 (16IM53), Q4b; SEE Mar/Apr 2023 (18IM53), Q3b (outline any one method of estimating process capability); SEE Nov/Dec 2019 (16IM53), Q3c (note on process capability study)	06	–	–

Process capability is the natural, inherent variability of a process once it is in statistical control, that is, once only chance causes remain. It is expressed as the natural tolerance 6σ (natural tolerance limits $\mu \pm 3\sigma$) and compared with the specification width through $C_p = (USL - LSL)/6\sigma$ and C_{pk}.

It can be estimated by three methods: the histogram or probability plot, the control chart method (preferred, since it confirms statistical control before capability is estimated) and designed experiments. The control chart method using the average range is explained below.

Procedure using the average range method (control chart method)

(1) Select the critical quality characteristic and a rational subgroup size n (usually 4 or 5).
(2) Collect at least 20 to 25 subgroups from the process operating under normal conditions.
(3) Compute $\bar{x}_i$ and R_i for every subgroup, then $\bar{\bar{x}}$ and $\bar{R}$.
(4) Construct $\bar{x}$ and R charts with limits $\bar{\bar{x}} \pm A_2\bar{R}$, $D_3\bar{R}$, $D_4\bar{R}$; remove subgroups with assignable causes and revise until the process is in control.
(5) Estimate the process standard deviation $\hat{\sigma} = \bar{R}/d_2$ (d_2 from the table for the subgroup size).
(6) Process capability $= 6\hat{\sigma}$; natural tolerance limits $= \bar{\bar{x}} \pm 3\hat{\sigma}$.
(7) Compare with the specifications: $C_p = (USL - LSL)/6\hat{\sigma}$, $C_{pk} = \min(USL - \bar{\bar{x}},\ \bar{\bar{x}} - LSL)/3\hat{\sigma}$, and estimate the percentage outside the specifications using the normal table.
(8) Decide: if $C_p \geq 1.33$ the process is capable; if $C_p < 1$ reduce variability, change the process or widen the tolerance; if $C_{pk} < C_p$ re-centre the process.

If an s chart is used instead of the R chart, the standard deviation is estimated as $\hat{\sigma} = \bar{s}/c_4$.

37	What are the process capability indices C_p and C_{pk}? Explain their significance. Source: Unit II | SEE Feb/Mar 2026 (IM353IA), Q4b	06	2	3

Process capability index C_p: $C_p = (USL - LSL)/6\sigma$, with σ estimated from $\bar{R}/d_2$ or $\bar{s}/c_4$. It compares the width of the specification tolerance with the natural spread (6σ) of the process and assumes the process is centred on the target, so it measures potential capability only.

C_{pk}: $C_{pk} = \min(CPU, CPL)$ with $CPU = (USL - \mu)/3\sigma$ and $CPL = (\mu - LSL)/3\sigma$. It accounts for process centring by using whichever specification limit is closer to the process mean, and measures the actual, realised capability.

Significance

Value	Meaning
$C_p < 1.00$	Process spread is wider than the tolerance; not capable
$C_p = 1.00$	Spread exactly equals the tolerance; marginal (0.27% outside if centred)
$1.00 \le C_p < 1.33$	Capable, but with limited margin for drift
$C_p \ge 1.33$	Generally regarded as capable (common industry minimum)
$C_p \ge 2.00$	Six Sigma level capability
$C_{pk} \le C_p$	Always; equality only when the mean is exactly centred between the limits
$C_p \gg C_{pk}$	Centring problem: capable in spread but running off target; re-centre the process
$C_{pk} < 0$	The process mean has moved outside the specification limits

In practice both indices are reported together: C_p shows the potential capability if the process were perfectly centred and C_{pk} shows the actual capability. A low C_p calls for reduction of variability; a low C_{pk} with an adequate C_p calls for re-centring.

38	A stable, normally distributed process has the specification 3.5 ± 0.03. Given $\mu = 3.51$ and standard deviation $= 0.016$, determine C_p and C_{pk}. What conclusions can you draw about the capability of the process? Source: Unit II | SEE Jan/Feb 2021 (18IM53), Q4b	04	–	–

$$LSL = 3.47, \quad USL = 3.53, \quad \mu = 3.51, \quad \sigma = 0.016$$

$$C_p = \frac{USL - LSL}{6\sigma} = \frac{3.53 - 3.47}{6(0.016)} = \frac{0.06}{0.096} = 0.625$$

$$C_{pk} = \min\left(\frac{3.53 - 3.51}{3(0.016)}, \frac{3.51 - 3.47}{3(0.016)}\right) = \min(0.417,\ 0.833) = 0.417$$

Both indices are well below 1. The natural spread of the process (6σ = 0.096) is 1.6 times the tolerance (0.06), so the process is **not capable** even if centred; in addition the mean is 0.01 above the nominal 3.50, which lowers C_{pk} further. About 11.2% of the output will be outside the specifications (10.6% above the USL). The process variability must be reduced (or the process changed) and the mean re-centred at 3.50.

Answer: Cp = 0.625, Cpk = 0.417: the process is not capable.

39	What are the two most common types of control charts for attributes? Briefly explain their purpose. Source: Unit II | SEE Feb/Mar 2025 (IM353IA), Q4a	06	2	1

The two most common attribute control charts are the p chart (fraction nonconforming) and the c chart (number of nonconformities).

p chart (fraction nonconforming): tracks the proportion of nonconforming units in each sample, $\hat{p} = D/n$, on the basis of the binomial distribution, with limits $\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n}$ (a negative LCL is set to 0). Each unit is classified simply as conforming or nonconforming, and the sample size may vary. Purpose: to monitor and reduce the proportion nonconforming, for example in final inspection results, audit results and service-quality pass/fail metrics.

c chart (number of nonconformities): tracks the number of nonconformities (defects) found in a constant, fixed-size inspection unit, on the basis of the Poisson distribution, with limits $\bar{c} \pm 3\sqrt{\bar{c}}$. A single unit may have several nonconformities and still be usable. Purpose: to monitor the number of defects, for example paint blemishes per car body, weld defects per assembly, typographical errors per printed page or imperfections per bolt of fabric.

Their variants are the np chart (number nonconforming, constant n), the 100p chart (percent nonconforming) and the u chart (nonconformities per unit, varying inspection unit size).

40	What are the differences between (i) p chart and np chart, (ii) c chart and u chart? (Also distinguish p and u charts, and p and c charts.) Source: Unit II | SEE Fast Track Oct 2020 (16IM53), Q4b; SEE Jan/Feb 2021 (16IM53), Q3b; CIE-II, 16/12/2025 (c and u); SEE Nov/Dec 2019 (16IM53), Q5b (p and c)	08	3	3

(i) p chart versus np chart

Basis	p chart	np chart
Plots	Fraction nonconforming $\hat{p} = D/n$	Number (count) nonconforming $D = n\hat{p}$
Sample size	May vary (limits form a staircase)	Must be constant
Limits	$\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n}$	$n\bar{p} \pm 3\sqrt{n\bar{p}(1-\bar{p})}$
Use	Samples of different sizes; management reporting (the 100p chart shows the same fraction as a percentage)	Easier for shop-floor operators: a headcount of bad parts, no fraction calculation

(ii) c chart versus u chart

Basis	c chart	u chart
Plots	Raw count c of nonconformities in an inspection unit	Rate: average nonconformities per unit $u = c/n$
Inspection unit	Constant, fixed size (one car body, one sheet)	May vary from sample to sample (n = number of inspection units)
Limits	$\bar{c} \pm 3\sqrt{\bar{c}}$	$\bar{u} \pm 3\sqrt{\bar{u}/n}$ (staircase when n varies)
Distribution	Poisson	Poisson (per-unit rate)

p chart versus u chart, and p chart versus c chart

Basis	p chart (p, np, 100p family)	c chart / u chart (c, u family)
What is counted	Nonconforming units: each unit is simply conforming or nonconforming; the number of defects on a unit is not counted	Nonconformities (defects); one unit may have several and still be usable
Distribution	Binomial	Poisson
Requirement	p: sample size may vary	c: inspection unit constant; u: inspection unit may vary
Example	Proportion of rejected transformers	Number of surface flaws per transformer

Rule of thumb: a conforming or nonconforming unit points to the p, np or 100p family; a count of defects where a unit can have several points to the c or u family.

41 Lots of bushing screws from a single supplier were inspected 100 percent with the following results. | 10 | 3 | 3

Lot No.	Amount in lot	Number rejected
1	6,000	183
2	6,000	131
3	10,000	184
4	1,783	23
5	3,089	171
6	5,774	65
7	3,000	46
8	2,724	85

Draw an appropriate control chart for the data. What can you conclude on the state of control?

Source: Unit II | CIE-II, 07/01/2025; CIE-II, 16/12/2025; SEE Feb/Mar 2025 (IM353IA), Q3b

The lot sizes are different, so a **p chart with variable control limits** is appropriate: $\bar{p} = \sum D_i / \sum n_i$ and limits $\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n_i}$ for each lot.

$$\bar{p} = \frac{\sum D_i}{\sum n_i} = \frac{888}{38,370} = 0.02314$$

$$\text{e.g. lot 1: } \sigma_{\hat{p}} = \sqrt{\frac{0.02314(1-0.02314)}{6000}} = 0.00194, \quad \text{UCL} = 0.0290,\ \text{LCL} = 0.0173$$

Lot	n	Rejected	$\hat{p}$	$\sigma_{\hat{p}}$	LCL	UCL	Status
1	6,000	183	0.0305	0.00194	0.0173	0.0290	Above UCL
2	6,000	131	0.0218	0.00194	0.0173	0.0290	In control
3	10,000	184	0.0184	0.00150	0.0186	0.0277	Below LCL
4	1,783	23	0.0129	0.00356	0.0125	0.0338	In control
5	3,089	171	0.0554	0.00271	0.0150	0.0313	Above UCL
6	5,774	65	0.0113	0.00198	0.0172	0.0291	Below LCL
7	3,000	46	0.0153	0.00275	0.0149	0.0314	In control
8	2,724	85	0.0312	0.00288	0.0145	0.0318	In control

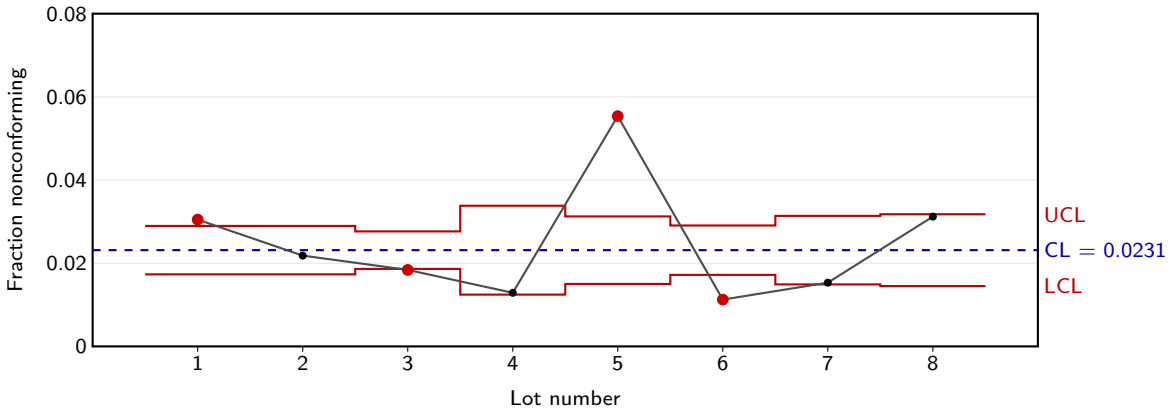


Figure 16: p chart with variable limits for bushing screws (CL = 0.0231)

Lots 1, 3, 5, 6 fall outside their control limits: lots 1 and 5 have far more rejects than expected, while lots 3 and 6 are below the LCL (better than the process average, possibly a genuine improvement or an inspection error). The supplier's process is **not in statistical control**; the quality of incoming lots varies far more than chance alone would explain. The supplier should investigate the assignable causes (especially for lot 5, with 5.5% rejects) before reliable limits can be set.

Answer: $\bar{p} = 0.0231$; lots 1, 3, 5, 6 are out of control, so the process is not in a state of statistical control.

42	Ten samples of 15 components are tested with the results below. Construct a p-chart and draw conclusions. Source: Unit II | SEE Jan/Feb 2021 (18IM53), Q4a	08	–	–

Sample	1	2	3	4	5	6	7	8	9	10
No. of defectives	2	1	3	2	3	1	0	1	0	1

$$\bar{p} = \frac{\sum D}{mn} = \frac{14}{10 \times 15} = 0.0933$$

$$\sigma_{\hat{p}} = \sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = \sqrt{\frac{0.0933(0.9067)}{15}} = 0.0751$$

$$\text{UCL} = 0.0933 + 3(0.0751) = 0.3187, \qquad \text{CL} = 0.0933$$

$$\text{LCL} = 0.0933 - 3(0.0751) = -0.1320 < 0 \Rightarrow 0$$

Sample	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
$\hat{p} = D/15$	0.133	0.067	0.200	0.133	0.200	0.067	0.000	0.067	0.000	0.067

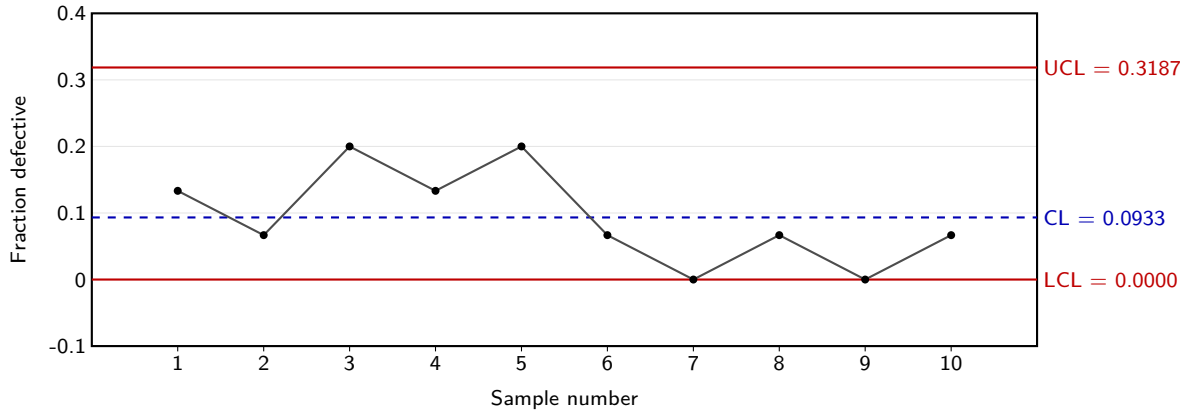


Figure 17: p chart for the components

The largest fraction defective is 3/15 = 0.200, below the UCL of 0.3187; no point is below the LCL (zero) and there is no unusual pattern. The process is **in statistical control** at an average fraction defective of 0.0933 (9.3%). Note that with n = 15 the chart is not sensitive to small shifts; larger samples would improve it.

Answer: CL = 0.0933, UCL = 0.3187, LCL = 0; the process is in control.

43	In a manufacturing unit of certain transformers the units are required to meet a number of specifications and are subjected to inspection daily. At the end of twenty days 230 units have been rejected out of 4150 produced and inspected. (i) Determine the 3-sigma control limits based on an average daily production of 200 units. (ii) Only one point has crossed the control limits: on that day 30 non-conforming items were found in 200 units inspected. (iii) What is the aimed value of p? (iv) What control limits would you recommend for future use? Source: Unit II | SEE Jan/Feb 2021 (16IM53), Q3a	10	–	–

(i) Three-sigma limits with n = 200

$$\bar{p} = \frac{230}{4150} = 0.0554, \qquad \sigma_{\hat{p}} = \sqrt{\frac{0.0554(1-0.0554)}{200}} = 0.0162$$

$$\text{UCL} = 0.0554 + 3(0.0162) = 0.1040, \qquad \text{LCL} = 0.0554 - 3(0.0162) = 0.0069$$

(ii) The out-of-control day

On that day $\hat{p} = 30/200 = 0.150$, which exceeds the UCL of 0.1040, so the process was out of control on that day. Assuming the assignable cause is found and corrected, that day's data are removed.

(iii) Aimed (revised) value of p

$$p_{\text{aim}} = \frac{230 - 30}{4150 - 200} = \frac{200}{3950} = 0.0506$$

(iv) Control limits for future use

$$\sigma_{\hat{p}} = \sqrt{\frac{0.0506(1 - 0.0506)}{200}} = 0.0155$$

$$\text{UCL} = 0.0506 + 3(0.0155) = 0.0971, \qquad \text{CL} = 0.0506, \qquad \text{LCL} = 0.0506 - 3(0.0155) = 0.0041$$

Answer: Trial limits 0.1040 / 0.0554 / 0.0069; revised (recommended) p = 0.0506 with UCL = 0.0971 and LCL = 0.0041 for a daily sample of 200.

44 The inspection results in a machine shop based on a sample size of 50 are given below. | 12 | – | –

Serial No.	Defectives	Serial No.	Defectives
1	6	11	3
2	3	12	7
3	1	13	1
4	2	14	15
5	12	15	4
6	6	16	18
7	4	17	3
8	7	18	2
9	1	19	6
10	8	20	7

(i) Calculate the control limits for the p-chart using 3σ limits. (ii) Plot the data and offer your comments on the behaviour of the process. (iii) What standard fraction defective do you recommend for the future period?

Source: Unit II | SEE Nov/Dec 2019 (16IM53), Q5a

(i) Control limits

$$\bar{p} = \frac{116}{20 \times 50} = 0.1160, \qquad \sigma_{\hat{p}} = \sqrt{\frac{0.1160(0.8840)}{50}} = 0.0453$$

$$\text{UCL} = 0.1160 + 3(0.0453) = 0.2519, \qquad \text{LCL} = 0.1160 - 3(0.0453) < 0 \Rightarrow 0$$

(ii) Plot and comments

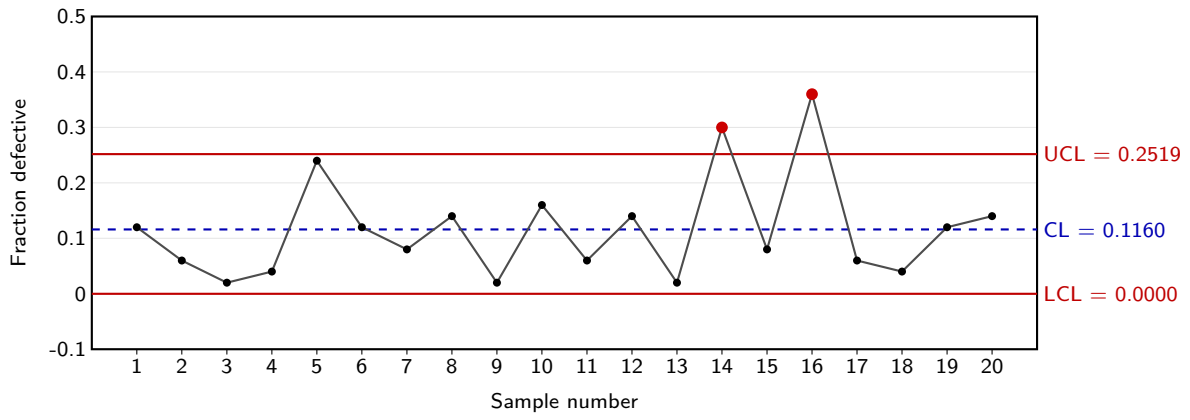


Figure 18: p chart for the machine shop (n = 50)

Samples 14, 16 (fraction defective 0.30 and 0.36) are above the UCL, so the process is **not in statistical control**; sample 5

(0.24) is also close to the limit. These samples indicate assignable causes (for example, tool breakage, a defective material lot or a new operator) that must be investigated and removed.

(iii) Standard fraction defective for the future

$$m = 20: \ \bar{p} = \frac{116}{1000} = 0.1160, \ \sigma_{\hat{p}} = 0.0453, \ \text{UCL} = 0.2519, \ \text{LCL} = 0.0000 \ \text{(out: 14, 16)}$$
$$m = 18: \ \bar{p} = \frac{83}{900} = 0.0922, \ \sigma_{\hat{p}} = 0.0409, \ \text{UCL} = 0.2150, \ \text{LCL} = 0.0000 \ \text{(out: 5)}$$
$$m = 17: \ \bar{p} = \frac{71}{850} = 0.0835, \ \sigma_{\hat{p}} = 0.0391, \ \text{UCL} = 0.2009, \ \text{LCL} = 0.0000 \ \text{(out: none)}$$

Answer: Trial limits: CL = 0.1160, UCL = 0.2519, LCL = 0. After removing samples 5, 14, 16, the recommended standard fraction defective is p = 0.0835, with UCL = 0.2009 and LCL = 0.

45	A process is being controlled with a fraction nonconforming control chart. The process average has been shown to be 0.07. Three-sigma control limits are used and the procedure calls for taking daily samples of 400 items. (i) Calculate the upper and lower control limits. (ii) If the process average should suddenly shift to 0.10, what is the probability that the shift would be detected on the first subsequent sample? (iii) What is the probability that the shift in part (ii) would be detected on the first or second sample taken after the shift? Source: Unit II | SEE Fast Track Oct 2020 (16IM53), Q4a; SEE Mar/Apr 2023 (18IM53), Q4a (parts i and ii)	08	–	–

(i) Control limits

$$\sigma_{\hat{p}} = \sqrt{\frac{0.07(0.93)}{400}} = 0.01276$$
$$\text{UCL} = 0.07 + 3(0.01276) = 0.1083, \qquad \text{CL} = 0.07, \qquad \text{LCL} = 0.07 - 3(0.01276) = 0.0317$$

(ii) Probability of detecting a shift to p = 0.10 on the first sample

$$\text{With } p = 0.10: \ \sigma_{\hat{p}} = \sqrt{\frac{0.10(0.90)}{400}} = 0.015$$
$$P(\text{detect}) = P(\hat{p} > \text{UCL}) + P(\hat{p} < \text{LCL}) = 1 - \Phi\left(\frac{0.1083 - 0.10}{0.015}\right) + \Phi\left(\frac{0.0317 - 0.10}{0.015}\right)$$
$$= 1 - \Phi(0.551) + \Phi(-4.55) = 0.2907$$

(iii) Detection on the first or second sample

$$P = 1 - (1 - 0.2907)^2 = 1 - (0.7093)^2 = 0.4968$$

Answer: UCL = 0.1083, LCL = 0.0317; P(detect on first sample) = 0.2907; P(detect by the second sample) = 0.4968.

46	An item is made in large lots and a sample of 200 items is drawn and given 100% inspection. The record sheet for the first 25 samples showed that a total of 75 items were defective. (i) Determine the trial control limits for the np-chart. (ii) Assuming that all the points fall within the control limits, what is the process average fraction defective? (iii) If p remains constant, what is the probability that the 26th sample will contain exactly 7 defectives? (iv) What is the probability that the 26th sample will contain 7 or more defectives? Source: Unit II | SEE Fast Track Jul 2019 (16IM53), Q4a	10	–	–

(i) Trial control limits for the np chart

$$\bar{p} = \frac{75}{25 \times 200} = 0.015, \qquad n\bar{p} = 200(0.015) = 3.0$$
$$\sigma = \sqrt{n\bar{p}(1-\bar{p})} = \sqrt{3.0(0.985)} = 1.7190$$
$$\text{UCL} = 3.0 + 3(1.7190) = 8.157, \qquad \text{CL} = 3.0, \qquad \text{LCL} = 3.0 - 3(1.7190) < 0 \Rightarrow 0$$

(ii) Process average fraction defective

$\bar{p}$ = 75/5000 = 0.015, i.e. 1.5% defective.

(iii) Probability of exactly 7 defectives in the 26th sample (binomial)

$$P(D = 7) = \binom{200}{7}(0.015)^7(0.985)^{193} = 0.0211$$

(iv) Probability of 7 or more defectives

$$P(D \geq 7) = 1 - \sum_{d=0}^{6}\binom{200}{d}(0.015)^d(0.985)^{200-d} = 0.0324$$

Answer: np chart: UCL = 8.157, CL = 3, LCL = 0; $\bar{p}$ = 0.015; P(D = 7) = 0.0211; P(D ≥ 7) = 0.0324.

47	A factory inspects different samples of a product, recording the number of defects in each sample. The sample size is constant, and the number of defects is provided below: Sample No. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 No. of defects | 5 | 8 | 6 | 4 | 7 | 9 | 5 | 6 (i) Draw an appropriate control chart for the data and evaluate the process. (ii) Plot the centreline, upper control limit (UCL) and lower control limit (LCL). Source: Unit II | CIE-II, 07/01/2025	10	2	CO2

The number of defects is counted in samples of constant size, so a **c chart** (Poisson) is appropriate.

$$\bar{c} = \frac{\sum c}{m} = \frac{50}{8} = 6.25$$
$$\text{UCL} = \bar{c} + 3\sqrt{\bar{c}} = 6.25 + 3\sqrt{6.25} = 13.75$$
$$\text{CL} = 6.25, \qquad \text{LCL} = \bar{c} - 3\sqrt{\bar{c}} = -1.25 < 0 \Rightarrow 0$$

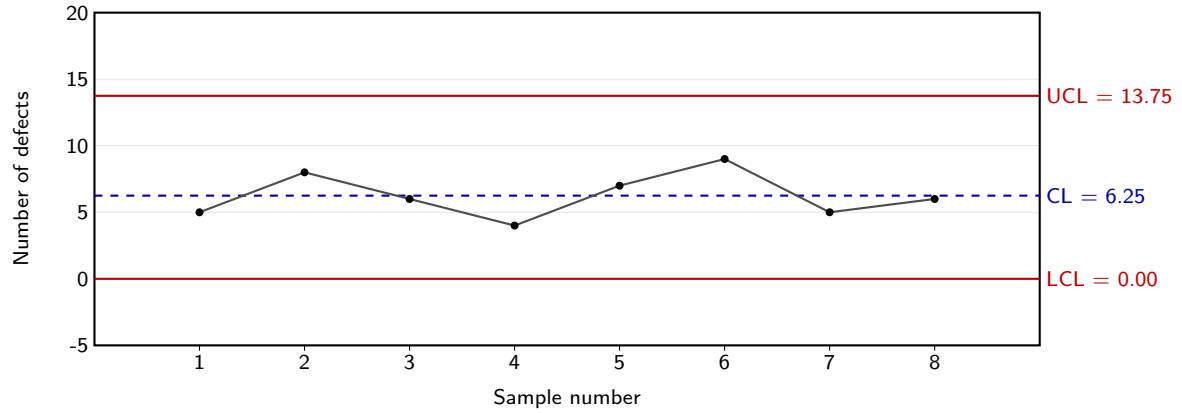


Figure 19: c chart for the number of defects

All eight points (4 to 9 defects) lie between the LCL (0) and the UCL (13.75) and show no run or trend. The process is **in statistical control** with an average of 6.25 defects per sample; any further reduction requires a change to the process itself (common causes).

Answer: c chart: CL = 6.25, UCL = 13.75, LCL = 0; the process is in control.

48 The following table gives the number of missing rivets at the aircraft final inspection: (10 | 4 | 4)

Sl No	Aircraft No.	Missing rivets	Sl No	Aircraft No.	Missing rivets
1	202	07	6	207	11
2	203	12	7	208	06
3	204	13	8	209	19
4	205	18	9	210	04
5	206	17	10	211	26

Establish an appropriate control chart for the purpose.

Source: Unit II | CIE-II, 16/12/2025

Each aircraft is one inspection unit and the number of missing rivets (nonconformities) is counted, so a **c chart** is appropriate.

$$\bar{c} = \frac{133}{10} = 13.30, \qquad \sqrt{\bar{c}} = 3.647$$

$$\text{UCL} = 13.30 + 3(3.647) = 24.24, \qquad \text{CL} = 13.30, \qquad \text{LCL} = 13.30 - 3(3.647) = 2.36$$

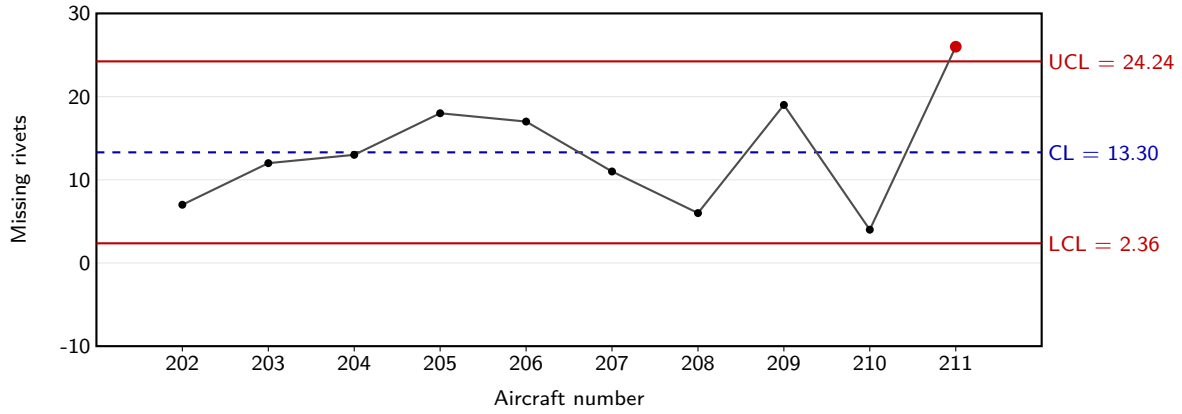


Figure 20: c chart for missing rivets

Aircraft 211, with 26 missing rivets, is above the UCL of 24.24; all other points are within the limits. Aircraft 211 indicates an assignable cause (for example, a new or untrained riveter, a faulty rivet gun or a supply problem) that must be investigated.

Revised limits for future use (aircraft 211 excluded)

$$\bar{c} = \frac{133 - 26}{9} = 11.89$$
$$\text{UCL} = 11.89 + 3\sqrt{11.89} = 22.23, \qquad \text{LCL} = 11.89 - 3\sqrt{11.89} = 1.54$$

Answer: Trial c chart: CL = 13.30, UCL = 24.24, LCL = 2.36 (aircraft 211 out of control). Recommended chart: CL = 11.89, UCL = 22.23, LCL = 1.54.

49	A control chart for nonconformities is maintained on a process producing desk calculators. The inspection unit is defined as two calculators. The average number of nonconformities per machine when the process is in control is estimated to be two. (i) Find the appropriate three-sigma control limits for this size of inspection unit. (ii) What is the probability of type I error for this control chart? Source: Unit II | SEE Mar/Apr 2023 (18IM53), Q4b	08	–	–

(i) Control limits

The average number of nonconformities is 2 per calculator; the inspection unit is two calculators, so $\bar{c} = 2 \times 2 = 4$ per inspection unit.

$$\text{UCL} = \bar{c} + 3\sqrt{\bar{c}} = 4 + 3(2) = 10, \qquad \text{CL} = 4, \qquad \text{LCL} = 4 - 3(2) = -2 \Rightarrow 0$$

(ii) Probability of type I error

$$\alpha = P(c > 10 \mid \lambda = 4) + P(c < 0) = 1 - P(c \le 10 \mid \lambda = 4)$$
$$= 1 - \sum_{x=0}^{10} \frac{e^{-4}4^x}{x!} = 1 - 0.99716 = 0.00284$$

Answer: UCL = 10, CL = 4, LCL = 0 per inspection unit of two calculators; type I error $\alpha = 0.0028$.

50	A cloth manufacturing company is maintaining 100% inspection of all the finished fabric. The past 8 months' data from the inspection indicated that the average number of defects per unit of fabric is 8. (i) Determine the 3σ control limits. (ii) With these control limits, what is the probability that the process is judged out of control when it really is not? Source: Unit II | SEE Fast Track Jul 2019 (16IM53), Q3b	04	–	–

(i) Three-sigma limits (c chart)

$$\bar{c} = 8, \qquad \text{UCL} = 8 + 3\sqrt{8} = 16.49, \qquad \text{CL} = 8, \qquad \text{LCL} = 8 - 3\sqrt{8} = -0.49 < 0 \Rightarrow 0$$

(ii) Probability of a false alarm (type I error)

$$\alpha = P(c > 16.49 \mid \lambda = 8) = 1 - P(c \le 16) = 1 - \sum_{x=0}^{16} \frac{e^{-8}8^x}{x!} = 1 - 0.99628 = 0.00372$$

Answer: UCL = 16.49, CL = 8, LCL = 0; probability of concluding out of control when in control = 0.0037.

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